Bitcoin's Required Return

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Abstract

What return should Bitcoin earn? Its history says 32% a year, asset pricing theory 6–18%, holders 14–22%. These are an extrapolation, a diversified investor's hurdle, and a survey answer. We build the model behind the third: what holders must require for Bitcoin's market risk, its volatility given their concentration in it, and its risk of death, from 520 deaths among 1,815 cryptoassets. It gives 15% a year today (90% range 12–18%), matching the survey answers, and far more early on; realized returns exceeded even that, an excess no measured risk explains, best read as adoption news and survivor selection.

Keywords: Bitcoin; cryptoassets; required return; expected return; survival analysis; limited participation; marginal holder; portfolio construction.

1. INTRODUCTION

Suppose you must write down one number: the return Bitcoin should be expected to earn, per year, over the next five years. Three candidate answers are on the table, and they disagree by a factor of three.

The first comes from Bitcoin's own history. A simple rule, the exponent of Bitcoin's power-law price trend, about 5.6 (Burger 2019; Papadogiannis 2026c), divided by Bitcoin's age in years, turned out to be the only candidate that forecast the returns that actually followed without bias over 2012–2026 (Papadogiannis 2026c). Applied today it says about 32% a year.[1] The second comes from standard asset pricing. The CAPM asks what Bitcoin's risk would cost a diversified investor: the ten-year Treasury yield of 5%, plus Bitcoin's beta to the stock market times an equity premium of 4% (6% as a high case). Because only a small part of Bitcoin's movement is shared with the stock market the answer is modest in most windows, but it depends on the window over which beta is estimated: 0.33 over the last four years, 1.06 since 2016, and 2.12 since 2012, giving 6–7%, 9–11%, and 13–18% (Papadogiannis 2026c); the 6–18% range is the spread across estimation windows, its upper end owed to the early, segmented years. The third comes from

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[1]Units, once and for all: the age rule's instantaneous (arithmetic mean) drift at age A is $\beta/A \approx 32\%$ at $A = 17.4$; its five-year-average form, used in the tables, is $(1+5/A)^{\beta/5} - 1 \approx 33\%$. The compound (geometric) counterpart is lower by roughly half the variance, $\sigma^2/2$, and so depends on the volatility assumed: a drag of about 4.5 points at 30% volatility and 10 points at 45%, which is why the compound figure is a range, 22–28%, and the arithmetic figure a single number. Section 7 says which form each test uses.

asking people. In a large US household survey, those who hold crypto said they expected 22% over the next year in 2021 and 13.8% in 2025 (Candia, Coibion, Gorodnichenko, and Weber 2026); 14–22% is the spread between the two dates. Neither range is a confidence interval; each is one quantity read in several ways.

These are not three estimates of one quantity; the disagreement begins with using one word for three things. The first is an *expected return*: a property of Bitcoin's price process, checkable only against what it later earns. The second is a *normative required return*: what an investor holding the CAPM's diversified portfolio ought to demand for Bitcoin's risk, whether or not any such investor holds it. The third is the *required return of the people who actually hold Bitcoin*, revealed by their portfolios and stated expectations; it is the one the price must satisfy in equilibrium, because they are the ones who must be willing to keep holding. The companion paper that documented the first number (Papadogiannis 2026c) closed by posing exactly this puzzle: is the gap between what Bitcoin delivered and what any model says it should deliver *compensation for risks that never materialized*, or *the mark of an asset the market underpriced*? Realized returns alone cannot say, because Bitcoin is one asset with one history, studied precisely because it survived.

This paper answers by measuring the two things the time series cannot identify. The logic is simple. Bitcoin pays no dividend or coupon, so it is worth holding only if its price is expected to rise by enough to pay its holders for the risks they take. Three risks matter, and each has a price: the day-to-day volatility of the asset, which costs more the larger the share of her wealth a holder keeps in it; the systematic risk it shares with the stock market, a small part of its total risk (Bitcoin's correlation with equities is low; its beta can still exceed one, as it does in the longer windows above, only because its own volatility is many times the market's, and the return it earns for that, beta times the equity premium, is a few points); and the risk that the asset dies outright, through a ban, a protocol failure, or a collapse of the kind that ended Terra and FTX. Add the three to the risk-free rate and you have the return the holder requires. The first two are measurable from Bitcoin's own prices and from surveys of who holds it. The third cannot be measured from Bitcoin, which has not died; it can only be measured from the deaths of other cryptoassets, as a life insurer prices one person from mortality tables of many. That is why a paper about Bitcoin's return spends a third of its length on the deaths of other coins.

First, we measure the risks that never materialized, by looking at the assets for which they did. We build a panel of 1,815 cryptoassets from 2010 to 2026, including 520 that died, and estimate the asset class's mortality curves as a demographer would. (Earlier descriptive studies of "coin death" exist, notably Feder, Gandal, Hamrick, and Moore (2018) and Grobys and Sapkota (2020); new here is a curated, pricing-oriented census: death types, dates, recovery fractions, size- and elite-conditional hazards, and the Lindy test.) Two results matter. The popular "Lindy" intuition, that the longer an asset has survived the safer it is, with hazard falling like one over age, is rejected: the death rate of cryptoassets *rises* with age. And the hazard that matters for Bitcoin, the death rate of assets that reached the top twenty, is small: 3.5% a year by any cause, and 0.30% a year for the sudden, value-destroying collapses (Terra's LUNA and UST and FTX's FTT, all in 2022). Priced at near-total loss, survival risk is worth a few points of expected return; it cannot explain a gap of roughly 27 points (the age rule's 32% less the $\sim$5% risk-free rate).

Second, we derive the return Bitcoin's actual owners must require, from the risks they bear and their own portfolios. The investor whose willingness to hold sets Bitcoin's price is not the diversified investor of the CAPM. The average crypto-owning household held 19% of its financial assets in crypto in 2021; option prices from the same era imply the marginal trader was closer to all-in. For such an investor,

elementary portfolio mathematics ties the return Bitcoin must promise to three observable quantities: volatility, portfolio weight, and risk aversion. We call the resulting formula the marginal-holder condition (Section 2, equation 1): risk aversion times variance times portfolio weight, plus small market and mortality terms. The route is Merton's (1987) limited-participation pricing with every input measured, not a factor regression and not a consumption-based model, neither of which can see the investor who actually clears this market. It is consistent with what holders said they expected in 2021 *and* in 2025 at ordinary levels of risk aversion (1.7 and 2.8). The required return was enormous when Bitcoin's volatility was 120% and its holders were concentrated, and it has fallen, term by measurable term, to about 15% today (90% range 12–18%).

With both measurements in hand, Bitcoin's history reads differently. Realized returns decompose into the required return of the marginal holder plus a *wedge*: the part the holder was not promised and could not have demanded. The wedge was on the order of 100 points a year in 2012–2015, near zero by 2019, and negative after late 2021, when holders required 21–58% and received about 13%. The age rule, we show, is the smooth one-over-age curve drawn through required-plus-wedge; it backtested as unbiased because the two components happened to decay together over the sample, and it has drifted high since 2020 (a drift its own out-of-sample record already shows) because the wedge has run out and changed sign.

What this means in practice. In plain terms, the findings are these. Bitcoin's price has to rise by about 15% a year to keep its current holders willing to hold it; that is the return its risk justifies today, and it has fallen from well over 100% a year when Bitcoin was volatile and held by a few concentrated enthusiasts. The much higher returns of Bitcoin's past were not a reward for risk but the market catching up with adoption it had not foreseen, and that catch-up ended around 2021. For an allocator, three things follow. Extrapolating Bitcoin's history (the 32%) overstates its prospective return by about half. A diversified investor who holds a small position collects a premium of roughly 4–7 points a year that exists only because Bitcoin's holder base is still concentrated, and that premium will shrink as the holder base broadens. And the position size the evidence supports for an investor who accepts the 15% is on the order of 10% of financial wealth for a moderately risk-averse investor, far below what Bitcoin's history would suggest; for smaller cryptoassets the same arithmetic says their holders have never been paid for the risks they carry.

The paper's contributions, in order: (i) mortality curves for cryptoassets built for pricing, with the Lindy hypothesis tested and rejected and the top-asset hazard measured; (ii) a required-return estimate for Bitcoin built on Merton's (1987) limited-participation condition from its holders' measured portfolios, their stated expectations, and option-implied bounds: three sources, one equation; (iii) a decomposition of Bitcoin's realized drift that explains both why the age rule worked and why it has begun to fail; and (iv) a dated, falsifiable forecast: about 15% a year, with a 90% range of 12–18%, for September 2026–2031, against the age rule's 32%. As checks, the same condition is applied to Ether, to smaller cryptoassets, and (Appendix C) to gold, where it reproduces a known result.

Section 2 sets out the marginal-holder condition, in words first. Section 3 describes the data. Section 4 presents the graveyard. Section 5 measures what holders require. Section 6 rereads Bitcoin's history as required return plus wedge. Section 7 states the forecast and what would falsify it; Section 8 applies the same pricing to Ether and to smaller cryptoassets; Section 9 translates the results for an allocator and compares Bitcoin with equities; Section 10 lists limitations; Section 11 concludes. Appendix C extends the condition to gold and to forward-looking hazards.

2. TWO RETURNS, AND THE PERSON WHO SETS THE PRICE

Everything in this paper runs through one equation, so we build it slowly and then display it.

Expected versus required. An asset's *expected return* is a forecast: what it will earn on average. Its *required return* is a hurdle: what an investor needs it to earn to be willing to hold it, given its risk. The first is positive, a statement about the price process that only realized returns can check, and they check it badly: Elton (1999) shows that realized returns are so noisy a proxy that decades of data can fail to distinguish competing values. The second comes in two forms that are often run together. The *market's* required return is what the marginal holder actually demands, a positive fact about people rather than prices, revealed by portfolio choices (Giglio, Maggiori, Stroebel, and Utkus 2021), stated expectations (Greenwood and Shleifer 2014), and option prices (Martin 2017). The *normative* required return is what an investor with stated preferences ought to demand, derived from a model; the CAPM number is of this kind (Sharpe 1964; Lintner 1965): a hurdle, not a description of anyone. In equilibrium the expected return equals the market's required return: the price adjusts until the forecast equals what the marginal holder demands. That equality is the whole content of "the market is in equilibrium"; it is why Cochrane (2011) can treat expected returns and discount rates as one object, and testing it is what empirical asset pricing does (Fama and French 2004). Forecasters work on the left side of the equality and financial economists on the right; the subject lives in the gap. For an asset whose holders are still learning what it is, the gap can stay open for years, because learning makes realized returns predictable in ways no required return explains (Lewellen and Shanken 2002; Pástor and Veronesi 2009a); the separation of the two returns for valuation is developed in Papadogiannis (2026a, 2026b).

In these terms the age rule estimates the expected return, the CAPM figures are normative required returns, and holders' surveyed expectations are evidence on the market's required return. The surveyed expectations are a number without a model behind it; this paper builds the model, deriving the market's required return for Bitcoin from the risks its holders bear, with every input measured, and checking the result against what holders say and what options imply. It uses the normative form only for the diversified allocator of Section 9, and then asks what remains of the expected return once the required return is subtracted; that remainder, which Section 6 calls the wedge, is the paper's estimate of the gap.

Whose hurdle? Here is the step most discussions skip. An asset has as many required returns as it has potential investors, but only one of them is reflected in the price: the hurdle of the *marginal holder*, the investor who is just indifferent between holding a little more and a little less. If a pension fund would be happy to hold Bitcoin at an expected 8%, but pension funds are not allowed to hold it, their 8% is irrelevant to the price. The price is set by whoever actually clears the market.

What risk aversion means. Everything rests on one assumption about people: an extra euro is worth less to you when you are richer. Risk aversion, γ, is the rate at which that happens: if your wealth rises by 1%, the value you place on one more euro falls by γ%. With $\gamma = 0$ a euro is a euro whatever your wealth, so you would take any fair bet; with $\gamma = 2$ each extra euro is worth 2% less for every 1% richer you become; with $\gamma = 4$, 4% less. The larger γ, the more a loss hurts relative to how much an equal gain helps. This is the Arrow–Pratt coefficient of relative risk aversion, and "constant" relative risk aversion means it does not change with wealth. For gambles the consequence is direct: a fair gamble leaves such a person worse off, and the payment she needs to accept it is about $\gamma/2$ times the gamble's variance, as a share of wealth. So $\gamma = 2$ means paying roughly 1% of wealth to avoid a coin flip over ± 10% of wealth; $\gamma = 1$ (log utility) pays

0.5%, $\gamma = 4$ pays 2%. The range 1 to 4 is not a convention but where estimates keep landing: about 2 from US household asset holdings (Friend and Blume 1975), below roughly 2 from labor-supply behavior (Chetty 2006), about 1.7–2 from Italian household panels, which also support the constant-relative form (Chiappori and Paiella 2011), and small single digits in the macro calibrations that followed Mehra and Prescott (1985); 2–3 is the workhorse range.

The marginal holder's arithmetic. Suppose this investor holds a fraction w of her wealth in an asset with annual return volatility σ whose return is essentially uncorrelated with the rest of her portfolio, a fair description of Bitcoin, whose correlation with equities has stayed below 0.2 for most of its life (Liu and Tsyvinski 2021). Should she hold a little more? The benefit is the asset's expected return above the risk-free rate, the *excess return*, since the euro could otherwise sit in a Treasury bill; that risk-free rate is the baseline for every return in this paper. The cost is the extra risk added to her portfolio, which, because the rest of the portfolio is uncorrelated with the asset, is proportional to how much she already holds, w, times its variance, σ^2. Her price for bearing risk is γ, so the cost in return units is $\gamma\sigma^2 w$. (The $\gamma/2$ of the gamble example has not gone missing: it prices the whole position, whose variance as a share of wealth is $\sigma^2 w^2$, at a total cost of $\gamma\sigma^2 w^2/2$; the decision here is about one more euro, so the relevant cost is the marginal one, the derivative of that total with respect to w, which is $\gamma\sigma^2 w$.) She stops adding when benefit equals cost: expected excess return $= \gamma\sigma^2 w$. The three multipliers are symmetric: halving her weight lowers the return she requires exactly as much as halving her risk aversion or the variance would, which is why a dilution of the holder base, as in the ETF era, lowers the required return just as a fall in volatility does. With the inputs measured later, Bitcoin's current volatility $\sigma = 0.42$ (Section 5), so $\sigma^2 \approx 0.18$, and the surveyed share of crypto in holders' financial assets, $w = 0.19$ in 2021 (Section 5), a holder of middling risk aversion $\gamma = 2$ requires $0.18 \times 0.19 \times 2 \approx 6.7$ points a year above the risk-free rate for the volatility alone; with $\gamma = 4$, twice that; a holder with only 5% of her wealth in Bitcoin, 1.8 points. Read one way this is a sizing rule: given the return you expect and your γ, hold $w =$ excess return$/(\gamma\sigma^2)$. Read the other way, which is what this paper does, an observed weight and an observed volatility reveal the excess return the holder must have been requiring. Adding the two risks this simplest version leaves out, market risk and sudden death, gives the central equation of the paper, which we call the **marginal-holder condition**. It is Merton's (1969) portfolio first-order condition, used as a pricing statement for an asset that only some investors hold as in Merton (1987), with the compensation for a jump to ruin from Merton (1976):

$$\boxed{\mu - r_f \;=\; \underbrace{\beta\,\mathrm{ERP}}_{\text{market risk}} \;+\; \underbrace{\gamma\sigma^2 w}_{\text{concentration}} \;+\; \underbrace{\lambda\,(1-\delta)}_{\text{mortality}}} \tag{1}$$

Read the terms left to right. The left side is the asset's expected return over the risk-free rate. On the right: a small CAPM-style term for whatever market risk the asset carries (beta times the equity risk premium); the **concentration term**, γ times variance times the holder's own portfolio weight, which prices the risk she cannot diversify because she chooses, or is forced, to hold a lot of one thing; and a **mortality term**, the probability per year that the asset dies, λ, times the fraction of each euro lost when it does, $(1-\delta)$. (Every term is stated per euro held in the asset; the holder's portfolio as a whole loses $(1-\delta)$ times her weight w.)

The mortality term, carefully. The two symbols answer different questions. λ is the probability per year that the death event occurs. δ is the *recovery fraction*, the share of her money a holder keeps when it does, so $(1-\delta)$ is the share she loses. They must be multiplied because dying and losing everything are not

the same thing. When TerraUSD collapsed, holders kept about two cents on the dollar ($\delta \approx 0.02$). When BUSD was wound down under regulatory order, a death by any institutional definition, holders redeemed at par: $\delta \approx 1$, and nobody lost a cent. When a token merely migrates tickers, $\delta = 1$ exactly, which is why Appendix A does not count migrations as deaths. The product $\lambda(1-\delta)$ is the *expected loss rate from mortality*, and it belongs in the drift because over a short interval dt the holder earns the surviving drift $\mu\,dt$ with probability $1-\lambda\,dt$ and loses $(1-\delta)$ with probability $\lambda\,dt$, so her expected return counting both outcomes is $\mu\,dt - \lambda(1-\delta)\,dt$; setting that equal to the required $r\,dt$ gives $\mu = r + \lambda(1-\delta)$. The price must climb, while the asset lives, by exactly the expected death loss, an actuarially fair insurance premium paid in drift, so that averaged over survival and death the holder earns what she requires. A 3% annual death probability with 95% lost on death costs $0.03 \times 0.95 \approx 2.9$ points a year. This is also the precise sense in which a backtest run on a surviving asset overstates the expected return: it measures μ, the drift on the paths where death did not happen, while the unconditional expected return is $\mu - \lambda(1-\delta)$; the crashes that would have pulled the average down are missing from the sample by construction (Brown, Goetzmann, and Ross 1995).

What equation (1) is, and is not. Nothing in equation (1) is new, and that is deliberate. The concentration term is the first-order condition of textbook portfolio choice (Merton 1969), and its use as a pricing statement under limited participation is Merton (1987): when only a fraction of investors hold an asset, those who do must be paid for the idiosyncratic variance they absorb, in proportion to how much each absorbs. The mortality term is the standard jump compensation (Merton 1976) read through the survival conditioning of Brown, Goetzmann, and Ross (1995), and a term of the same form, hazard times loss, appears wherever a portfolio problem includes a jump to ruin, from event-risk allocation (Liu, Longstaff, and Pan 2003; Das and Uppal 2004) to the rare-disaster models that followed Barro (2006). The beta term is the CAPM. What is ours is the assembly and, above all, the mapping: each term matched to something observable for this asset class, the holder's weight to a survey, the death hazard to a census of the asset's relatives, the whole to stated expectations and option prices. To our knowledge that mapping has not been written down for cryptoassets; the contribution is not the equation but its inputs. The mechanism the concentration term describes has a name and an empirical record. Merton (1987) drew the conclusion himself: assets held by few carry a higher cost of capital, and it falls as recognition broadens, the *investor recognition hypothesis*. Levy (1978) had derived the pricing of idiosyncratic variance by investors who hold few securities a decade earlier, and Errunza and Losq (1985) the same form for securities that foreign investors cannot hold. The hypothesis has been tested for individual stocks, where broadening the shareholder base raises the price and lowers subsequent returns (Kadlec and McConnell 1994; Foerster and Karolyi 1999; Amihud, Mendelson, and Uno 1999; Lehavy and Sloan 2008; Bodnaruk and Östberg 2009), and for national markets opening to foreign capital (Bekaert and Harvey 2000; Chari and Henry 2004). Pástor and Veronesi (2009b) give it a lifecycle: a new technology's risk is idiosyncratic while it is small and held by few, and turns systematic as it is adopted. Ibbotson, Idzorek, Kaplan, and Xiong (2018) generalize it into a theory of popularity, in which hard-to-diversify assets earn a premium that popularity erodes. For cryptoassets the record is indirect: a broadening investor base is linked to idiosyncratic volatility (Izadyar and Zamani 2022), and institutional participation, the spot ETFs in particular, raised Bitcoin's correlation with equities (Copestake, Furceri, and Terracciano 2026), as index investment did for commodities (Tang and Xiong 2012). What the record lacks is the size of the mechanism for an asset class: every test above is a sign test on prices or correlations. Mea-

suring the level, with the holder weight from surveys and the death hazard from a census, is what this paper does. Three clarifications. First, equation (1) is an optimality condition that any equilibrium must satisfy, not an equilibrium model: we do not model supply, endogenize w, σ, or λ, or derive the price level. It complements rather than competes with the equilibrium models of Bitcoin's price (Biais, Bisière, Bouvard, Casamatta, and Menkveld 2023; Pagnotta 2022; Cong, Li, and Wang 2021), supplying the discount-rate input they leave to be estimated. Second, the condition is generic by design: the same line prices founder stock, private equity, and emerging-market equities before liberalization. Bitcoin enters through the values of the inputs, not the functional form, and through one data advantage no other asset class offers: a thousand-asset graveyard from which λ can be measured rather than assumed. The emerging-market integration literature (Bekaert and Harvey 2000; Henry 2000) documents the mechanism our forecast rests on: required returns are high while a market is segmented and compress when it opens to outside capital; the ETF era is Bitcoin's liberalization. Third, using the survey-average holder weight for w is not casual: in an equilibrium of holders with similar risk aversion, every unconstrained holder's condition binds at the same $\gamma\sigma^2 w$, so weights equalize across participants and the average participant's weight identifies the marginal one's. The identification assumes that weights are chosen optimally under beliefs that are on average correct; a weight inflated by over-optimism satisfies the same first-order condition but recovers a belief rather than a fair premium, and Section 6 explains how the wedge tells the two apart. Every empirical section measures one term: $\lambda(1-\delta)$ from the graveyard (Section 4); σ, w, and γ from prices, surveys, and options (Section 5). Section 6 then evaluates the condition date by date, Section 7 uses it as a forecast, Section 8 applies it to Ether and other coins, and Section 9 inverts it into a sizing rule.

A worked example. In 2021, Bitcoin's volatility was about 80% a year, so $\sigma^2 = 0.64$. Survey data (Section 5) show the average crypto-owning household held $w = 19\%$ of its financial assets in crypto. With $\gamma = 2$ the concentration term is $2 \times 0.64 \times 0.19 \approx 24$ points a year. Add the risk-free rate of the time, 1.5%, a beta term of about 2 points (a beta of 0.5 on a 4% equity premium), and 0.3 points of mortality (Section 4), and the marginal holder of 2021 required about 28% a year. Holders surveyed that year said they expected 22%, which the same arithmetic delivers at $\gamma \approx 1.6$, well inside the ordinary range. Holders *saying* they expected 22% were not being exuberant; they were reading their own first-order condition.

What makes each term fall. Volatility falls as markets deepen: Bitcoin's σ has gone from about 150% (2013) to about 40–50% (2026). The weight w falls as participation broadens: when the only people willing to hold an asset are enthusiasts, each must hold a lot of it; when ETFs let everyone hold a sliver, nobody has to. And λ falls, or at least the market's estimate of it falls, as the asset survives attacks it was feared not to. Each term is measurable, each has fallen, and the sum today is about 15% a year; Figure 6 in Section 6 shows the path of each.

3. DATA

The asset panel comes from the Coin Metrics community dataset (Coin Metrics 2026), the same source the companion backtest paper uses for Bitcoin and Ether prices, observed from July 2010 to May 24, 2026. After removing fiat exchange rates, tokenized equities, wrapped and staked duplicates, stablecoins, and trading-pair artifacts, the core universe is **1,815 cryptoassets**, contributing about 67,700 asset-months of exposure. The dataset carries full daily *trading volume* for essentially all of them, and full daily *prices* for the 139 historically significant assets; volume therefore carries the life-and-death timeline, and prices carry

the return calculations on the significant subset.

Birth. An asset is born, for our purposes, on the first day it shows a price or trading volume: the day it enters the investable, tracked universe. This is the clock an investor cares about.

Death. Death is harder, because cryptoassets rarely die cleanly. We use three mechanical definitions and then a curated layer:

1. **Feed death**: the data feed ends more than 180 days before the dataset does. This proxies delisting from tracked markets. 234 cases.

2. **Price death**: the price falls below 1% of its all-time high and never recovers (measurable only on the price-complete subset). 33 cases.

3. **Volume death**: trailing three-month volume falls below 1% of its historical peak *and* below an absolute floor of $250,000 a day, and never recovers. The absolute floor matters: without it, healthy mid-caps whose volume merely normalized after the 2021 mania would be falsely counted as dead. 438 cases, overlapping the other two definitions.

4. **Curation.** Two corrections are applied by hand, with the full audit trail in Appendix A. *Ticker migrations are not deaths*: when MATIC became POL, EOS became Vaulta, or Golem's GNT became GLM, holders received successor tokens at full value, so these events (seventeen in all) are removed. And *known catastrophes are dated correctly*: the free data tier hides the price history of several major assets, so the collapses of LUNA and UST are entered at May 12, 2022 and FTT at November 10, 2022 from the public record, rather than at whatever date a volume rule would eventually notice them.

The result: **520 deaths among 1,815 assets (29%)**, of which 8 are classified as sudden, value-destroying *jump-collapses* and the rest as gradual abandonment: projects that faded, lost their listings, and stopped trading.

The other inputs are: Bitcoin's monthly price, market capitalization, and count of addresses with a balance (Coin Metrics); US household crypto-ownership rates from the Federal Reserve's payment and household surveys and Pew (0.4% of adults in 2014, 1.2% in 2018, 9–12% in 2021, roughly 11–14% since); holders' portfolio shares and expectations from Candia, Coibion, Gorodnichenko, and Weber (2026), with Austrian household-finance data (OeNB 2025) as a European cross-check; ten-year Treasury yields; global household wealth from the UBS Global Wealth Report; option-implied figures from Deribit as reported in the literature and the DVOL index; spot-ETF and public-company Bitcoin holdings from issuer disclosures and corporate filings (as compiled by Bitcoin Treasuries 2026); and, for the gold comparison of Appendix C, LBMA gold prices and World Gold Council above-ground stock estimates.

One bias deserves a sentence up front. Coin Metrics adds an asset to its universe only once the asset has become significant, so thousands of micro-cap tokens that died young and obscure never enter the panel. This *understates* the death rate at young ages, which means it biases the test *against* our headline finding that hazard rises with age. The elite estimates, which are what Bitcoin's pricing needs, are unaffected: once an asset is in the top twenty it is fully observed from then on.

4. THE GRAVEYARD: HOW CRYPTOASSETS DIE

A *hazard rate* answers a simple question: of the assets alive at a given age, what fraction die within the next year? If 100 three-year-old coins are alive in January and 12 are dead by December, the hazard at age three is 12% a year. Estimating hazards across ages, with assets that are still alive treated correctly as "censored" (we know they lived *at least* this long), is standard survival analysis; the machinery is the same used for patients, firms, and light bulbs.

The Lindy question. The age rule's most natural economic defense was mortality: perhaps Bitcoin's enormous drift was fair payment for a risk of sudden death that declines as the asset ages. This is the "Lindy effect," under which remaining life expectancy grows with age, so the hazard falls like k divided by age. If that were true with k near 5.6, the age rule would simply be survival compensation, rationally priced, with equation (1)'s mortality term doing all the work. The hypothesis is attractive, parameter-free in shape, and testable; testing it is the point of building the graveyard.

It fails. Table 1 shows the death rate by age bucket under the curated definition.

Table 1: Mortality hazard by age, curated death definition (1,815 assets, 520 deaths, 2010–2026).

Age (years)	Deaths	Exposure (asset-yrs)	Hazard (per yr)
0.1–0.5	6	693	0.9%
0.5–1	37	770	4.8%
1–2	111	1,213	9.2%
2–3	106	905	11.7%
3–4	82	661	12.4%
4–5	78	488	16.0%
5–7	63	502	12.6%
7–9	33	198	16.7%
9–12	4	53	7.5%

Buckets begin at age 0.1 (the first weeks of a listing are definitionally alive) and the sparse 12–17 bucket (7 asset-years, no deaths, Bitcoin the main occupant) is omitted; total exposure across all ages is about 5,640 asset-years.

The hazard *rises* with age through at least age nine. Formally, a Poisson regression of monthly deaths on log age, with calendar-year fixed effects to absorb market-wide waves such as crypto winters and standard errors clustered by asset, gives an age elasticity of **+0.42 (s.e. 0.04)**: each doubling of age *raises* the death rate by roughly a third, holding the calendar year fixed. Lindy requires -1. The estimate has the wrong sign by eleven standard errors, and it is robust: $+0.44$ to $+0.56$ across the three mechanical death definitions, $+0.65$ when the suspicious cluster of mid-2025 feed terminations is excluded, and positive within the elite subsample alone. Equivalently, the slope of log survival against log age, which equals k if the hazard really were k/age, gives **between about 0.2 and 0.5** depending on the death definition, **against the 5.6** the age rule would need. A 5.6 would have implied that a three-year-old cryptoasset had roughly a 1-in-20,000 chance of reaching Bitcoin's current age; the measured elite hazard, held constant, implies a 14-year survival probability around 0.6. Cryptoassets age like small firms, with a liability of adolescence and obsolescence, not like Lindy classics (Figure 1).

Where the data end. Three caveats belong next to that result. First, in the oldest populated bucket (ages 9–12) the raw hazard falls back to 7.5%, on 4 deaths and 53 asset-years; the rise is established through

roughly age nine, the data's support ends near age twelve, and Bitcoin is 17.4. The Lindy rejection is therefore a statement about the asset class over its observed life span, not a measurement of Bitcoin's own current hazard, which is exactly why the pricing below uses the elite hazard *level* rather than an age extrapolation. Second, elite deaths cluster in calendar time (2022, 2024–25), with contagion episodes and delisting purges, not smoothly in age; the three jump-collapses in particular sit within six months of one another and were causally linked (Terra propagated through Celsius to FTX), so the Poisson confidence interval, which assumes independent events, understates the uncertainty. We treat 0.30% as the point estimate of a partly systematic process, which a diversified allocator may rationally charge for in part; Section 9 does so. Third, the 2024–25 delisting purge supplies 12 of the 36 elite deaths; treating every 2025 death as censoring lowers the all-cause elite hazard from 3.5% to 2.4% a year (CI 1.5–3.5%), so the mortality bound of $\leq$3 points used below is conservative under either reading.

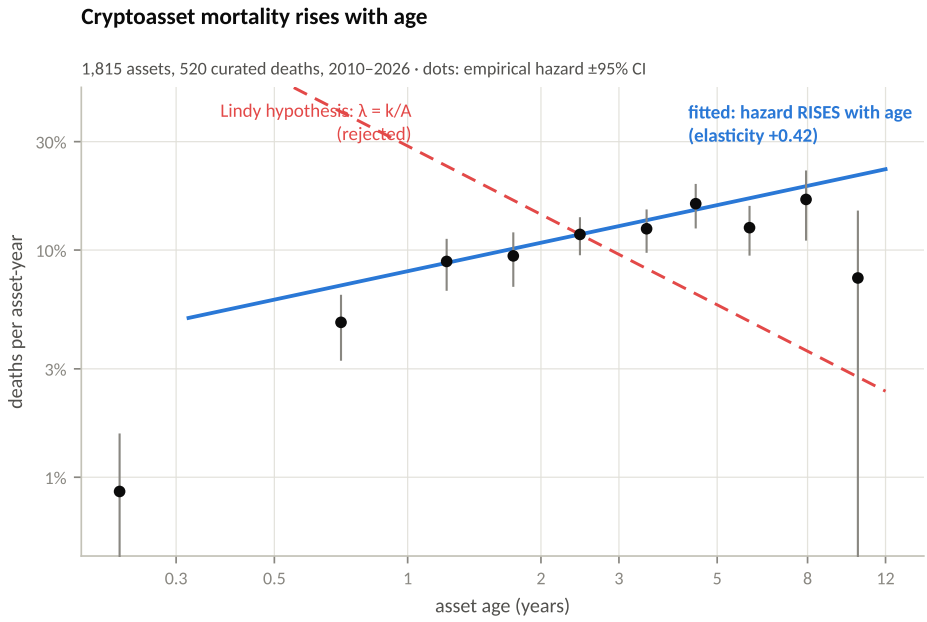


Figure 1: Mortality hazard versus age (log–log). The fitted line uses the period-fixed-effects elasticity of +0.42; the dashed line is the Lindy shape $\lambda = k/A$.

Size is what protects, not age. Sorting assets by peak trading volume (Table 2), the hazard falls monotonically and steeply with size.

Table 2: Mortality by size and for elite cohorts (curated deaths).

Size group	n	Hazard (per yr)
Smallest quartile	452	17.4%
Q2	451	13.0%
Q3	451	9.8%
Largest quartile	452	3.2%
Ever top-50 by volume	499	5.5% (CI 4.5–6.5%)
Ever top-20 by volume	**219**	**3.5% (CI 2.5–4.9%)**
Ever top-20, jump-collapse only	219	**0.30% (CI 0.06–0.86%)**

The last two lines are the ones Bitcoin's pricing needs. Among the 219 assets that ever reached the top twenty by monthly volume, 36 died over 1,017 asset-years of exposure, an all-cause rate of 3.5% a year, and only three of those deaths were the sudden, catastrophic kind that a jump-risk premium prices: LUNA and UST in May 2022 and FTT in November 2022, each a roughly total loss within days. None has occurred since. Excluding four contested cases, assets flagged dead by the drawdown rule but whose chains still operate (ICP, SUSHI, FLOW, NANO; Appendix A), the all-cause rate falls to 3.15% (CI 2.2–4.4%). The remaining elite deaths are slow fades: OMG, NEM, Bitcoin Gold and their like lost their listings over months or years, during which holders could exit at declining prices; losses of that kind live in an asset's volatility and drift, not in its jump hazard. The 520 dead assets are, in the taxonomy of the companion scarcity framework (Papadogiannis 2025), the "physically scarce but no longer desired" quadrant realized at scale: hard-coded supply caps outlived demand, and scarcity without desirability monetizes nothing (Figure 2; survival curves by size in Figure 3; the death timeline, with the 2022 cascade and the 2024–25 delisting purge, in Figure 4).

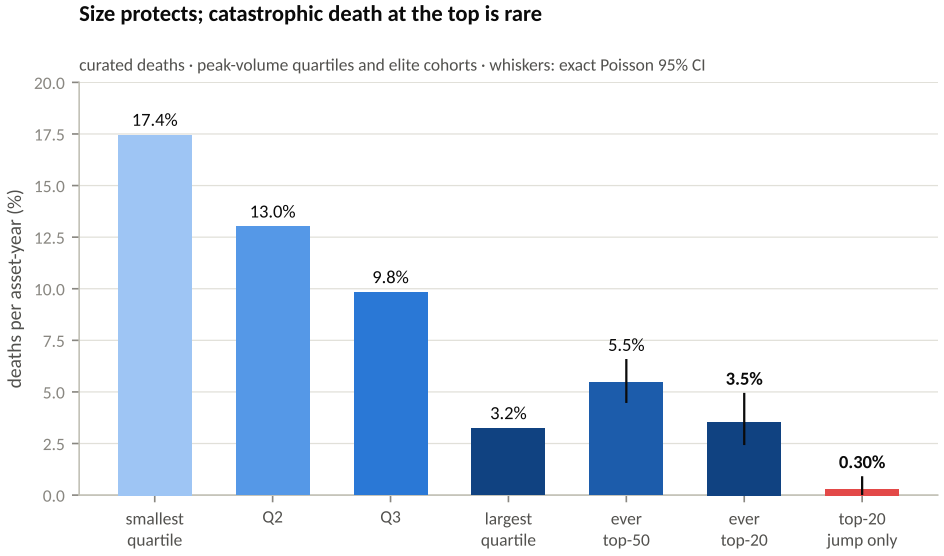


Figure 2: Hazard by peak-volume quartile and for the elite cohorts; whiskers are exact Poisson 95% intervals.

What this buys the pricing equation. Price the jump channel at near-total loss: $\lambda(1-\delta) \approx 0.3\% \times 0.95 \approx$ **0.3 points a year**. Treat every elite death, even the slow fades, as a jump at total loss and you still cannot get above $\approx$ **3 points**. The mortality term in equation (1) is real (2022 proved it can strike the top ten), but it is a few points, not twenty-seven. One caveat matters: no cross-section can measure Bitcoin's *own* hazard in 2010–2014, when it was the only asset of its kind and plausibly faced existential risk; our bounds apply from the point at which an elite class exists to measure, and Section 6 keeps that early window wide.

Risks that have never yet fired for anyone, the clearest being a quantum break of Bitcoin's signature scheme, are λ-events the condition covers by construction but the graveyard cannot price; Appendix C.3 sets out how the option-implied crash intensity referees them and how an investor adds a private view to the mortality term.

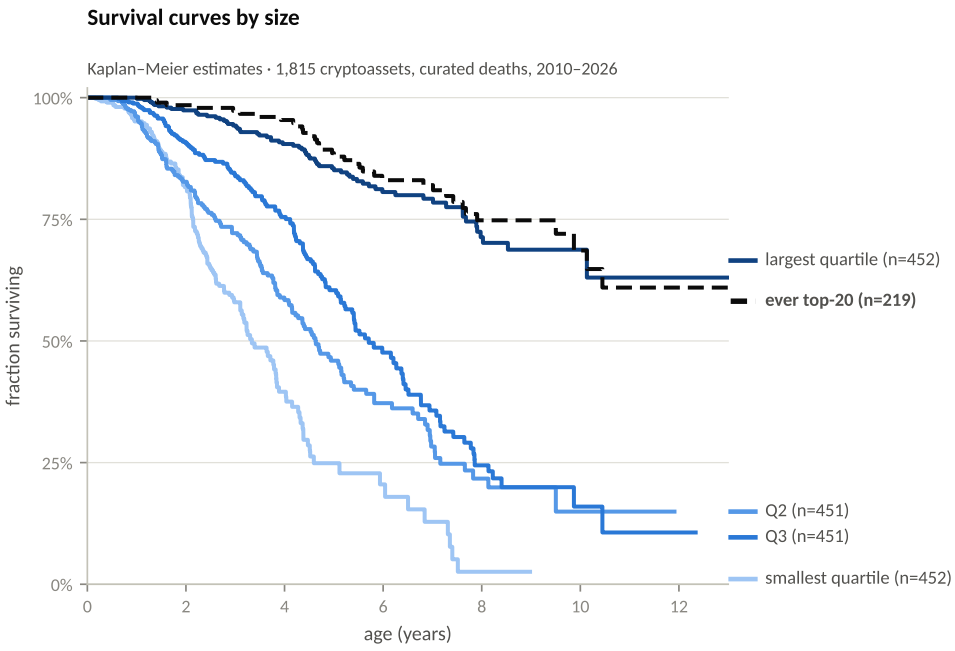


Figure 3: Kaplan–Meier survival curves by peak-volume quartile, with the ever-top-20 cohort (dashed). Half of the smallest quartile is dead by age 3.3; the largest quartile and the elite track each other and remain above 60% survival past age 10.

5. WHAT BITCOIN'S HOLDERS REQUIRE

The marginal-holder condition, equation (1), has four inputs on the right-hand side. The graveyard supplied λ. This section supplies the rest (volatility, the holder's portfolio weight, and risk aversion), assembles the required return, and checks it against two independent readings, holders' stated expectations and option prices; the three tell one story.

Source 1: holders' measured portfolios. The Cleveland Fed study of cryptocurrencies in household finance (Candia, Coibion, Gorodnichenko, and Weber 2026) reports the number the equation needs most: among US households that owned any crypto in 2021Q3, crypto made up **19% of financial assets on average**, with 40% of owners below 5% and about a fifth of owner-households above *half* their portfolio. A European cross-check points the same way: in the Austrian central bank's 2023 household survey, owners mostly held under 20% of financial assets in crypto, rarely above 20–25% (OeNB 2025). So the marginal holder of the early 2020s was not the diversified investor of the CAPM; she held a fifth of her wealth in the asset. For 2025–26 the surveys have not yet published a portfolio share, and the paper uses a range: 19% as the upper bound (the 2021 value, which the market-clearing back-cast of Section 6 also returns for survey-type holders in 2026) and 8% as the lower bound, obtained by adding the spot-ETF and public-company treasury holdings, about 2.5 million BTC by mid-2026, as a second class of holders who are not "crypto owners" in any survey and hold at about 1.5% of their wealth (Appendix D).

Source 2: holders' stated expectations, a consistency check. The same surveys asked holders what return they expected over the next year: 22% in 2021, 13.8% in 2025. Now run equation (1) forward at each date and ask what risk aversion γ makes it hold (Table 3).

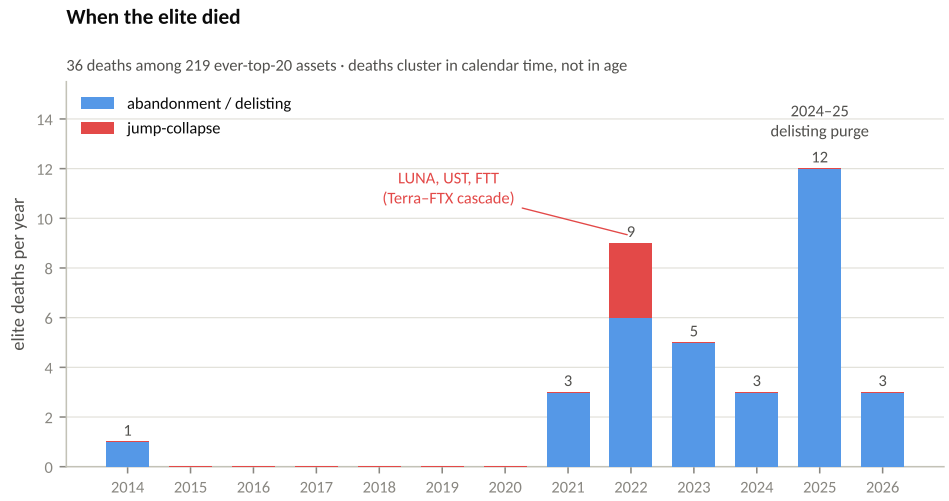


Figure 4: When the elite died: 36 deaths among 219 ever-top-20 assets, by year and type.

Table 3: Two survey waves, one equation.

	2021Q3	2025
Volatility σ (trailing)	∼80%	∼45%
Holder portfolio weight w	0.19 (measured)	0.08–0.19 (bounded)
Risk-free + beta term	∼3–4%	∼7–8%
Stated expected return	22%	13.8%
Implied γ	≈ 1.7	≈ 2.8

Implied γ solves equation (1) at $\sigma = 0.77$, $w = 0.19$, $r_f + \beta\text{ERP} = 3.5\%$ (2021) and $\sigma = 0.45$, $w = 0.11$, $r_f + \beta\text{ERP} = 7.5\%$ (2025), with the mortality term at 0.3 points. Across the stated input ranges the 2025 figure spans roughly 1.6–3.1.

Two clarifications. With one free γ per wave, each line can be made to hold by construction; this is a range-consistency check, not an overidentification test. Its content is that the implied γ lands both times inside the interval (roughly 1 to 4) that describes ordinary investors, not at the implausible values (0.2, or 15) that would expose the condition as irrelevant to how holders think. And the move from 1.7 to 2.8 is economically sensible rather than an embarrassment: as participation broadens from early enthusiasts toward the general investing public, the marginal holder should become more risk-averse; with one γ per wave this is a reading, not a test. Holders' expectations, so often dismissed as exuberance, are consistent with their own portfolio first-order condition, the beliefs-to-portfolios coherence that Giglio, Maggiori, Stroebel, and Utkus (2021) document for equity investors. This is the paper's quiet central exhibit: the 14–22% that "holders say they expect" was never a third, inconsistent number; it is the required return of the people who hold the asset, moving as its determinants move.

Source 3: option prices. Options give a holder-free reading. A classic result (Martin 2017) says the risk-neutral variance priced into options (the variance computed from option prices rather than from history) is a lower bound on the expected excess return demanded by an investor fully invested in the asset; in equation (1) "$\gamma w = 1$" describes a fully invested log-utility investor, whose required excess return is exactly the return variance, and Martin's result is that option prices put a floor at that variance. Deribit-

based estimates for 2017–2022 put that bound at **66–84% a year**, with Bitcoin implied volatility then running at 80–90% (Almeida, Grith, Miftachov, and Wang 2024; Deribit options, July 2017–December 2022). This is the concentration mechanism priced by the market itself, in the era when crypto markets were documentably segmented from mainstream capital, with large and persistent cross-market price gaps (Makarov and Schoar 2020), and the marginal trader really was close to all-in. Today Deribit's DVOL index (the 30-day option-implied volatility, a VIX for Bitcoin) has run between about 40 and 55, so the same all-in bound is 0.40^2–$0.55^2 \approx$ **16–30%**, and the actual marginal holder, who in the ETF era is no longer all-in but holds w of 8–19%, requires $\gamma\sigma^2 w \approx$ **2–9 points** above the risk-free-plus-beta floor.

Putting the 2026 number together. Risk-free 5.0%; beta term 1.3–4.2 points (beta 0.33–1.05 on a 4% equity premium: the four-year and since-2016 estimates; the since-2012 beta of 2.12 is dominated by the segmented early years and is not the beta a holder faces today); concentration 2–9 points ($\gamma \in [1.7, 2.8]$, $\sigma = 42\%$, $w \in [0.08, 0.19]$); mortality 0.3–3 points. Summing the extremes gives 8.6% to 21.2% a year, an *envelope* that we round to 9–22%. But an envelope puts every uncertain input at its extreme at once, and the inputs are independent: the lowest plausible beta has nothing to do with the lowest plausible risk aversion or holder weight. Treating each input as uniformly uncertain over its stated range and drawing them independently (Appendix B.5), the required return has a median of **about 15% a year**, a central 68% interval of 13–17%, and a **90% interval of 12–18%**; the corners of the envelope are reached with negligible probability. The holder weight contributes most of the width (about 4 points on its own), beta, mortality, and risk aversion roughly 3 points each. The age rule, for contrast, says 32% with a precision that is apparent rather than real: a one-parameter fit returns a point by construction, and in the companion paper's backtest that point missed the realized five-year return by amounts spread over tens of points. The interval here describes parameter uncertainty in the required return; the realized five-year outcome will in addition carry diffusive noise of roughly ± 19 points on the compound return at one standard deviation, which no forecast of a 42%-volatility asset can remove. The two cannot both be the expected return, unless the difference is something other than a required premium. Section 6 shows that historically it was exactly that: a wedge.

6. REREADING BITCOIN'S HISTORY: REQUIRED RETURN PLUS WEDGE

Write each period's realized return as the sum of two parts:

$$\text{realized} = \underbrace{\mu^{req}(t)}_{\text{what holders could demand}} + \underbrace{\varepsilon(t)}_{\text{the wedge: what nobody was promised}} . \tag{2}$$

The required part, $\mu^{req}(t)$, is equation (1) evaluated date by date from the measured ingredients: trailing volatility from the price series; the holder weight w bounded between a market-clearing back-cast and the all-in trader the option bounds describe; the risk-free rate of the date; and the beta and mortality terms held at their stated ranges throughout, since neither has a defensible time path (the point is discussed with Figure 6). The wedge is what is left. The back-cast generates the whole pre-2021 lower band, so it deserves plain words; Appendix D shows each step graphically. Market clearing requires that Bitcoin's market capitalization equal the wealth its holders have placed in it: cap $= \bar{w}\times$ (participating wealth). We anchor participating wealth in 2021Q3, the one date where both sides are measured (capitalization divided by the surveyed 19%), and move it backward and forward in proportion to the number of participants times aver-

age wealth per participant. The participant count is the share of US adults holding crypto in the household surveys of Section 3 (0.4% in 2014, 1.2% in 2018, 11% in 2021Q3, 12–13% since), interpolated log-linearly between survey dates, with the count of addresses holding a balance used only before the first survey; wealth per participant is a ramp from a quarter of its 2021 level in 2012, when holders were students and hobbyists, to 1.4 times by 2026, when they include institutions. From 2024 the spot ETFs and public-company treasuries add a second channel of holders who are not "crypto owners" in any household survey and hold small weights; the lower edge of the band counts their roughly 2.5 million BTC (mid-2026) as held at 1.5% weights, the upper edge does not count them. Dividing capitalization by that path gives the implied average holder weight at each date: about 3% in 2013, 17% in 2017, 18% in 2021 (the anchor), and in 2026 between 8% with the institutional channel and 21% without it. Read literally, the low early values say that *if* early Bitcoin wealth had been spread across survey-style households, each held a small position and required little; the all-in alternative reads the same capitalization as concentrated in few fully committed hands who required a great deal. The band is exactly the interval between those readings. Table 4 does this at two-year intervals; Figure 5 shows the full monthly series.

Table 4: Required return versus realized drift versus the age rule.

Year	Age	Volatility	Required (band)	Realized next-5y CAGR	Age rule (5-yr form)
2013	4.5	151%	17–231%	135%	132%
2015	6.5	115%	12–135%	103%	90%
2017	8.5	80%	22–67%	51%	68%
2019	10.5	90%	37–82%	42%	55%
2021	12.5	83%	21–58%	∼13% (2021Q3 to May 2026)	46%
2023	14.5	66%	12–24%	–	40%
2026	17.4	42%	9–22%	–	33%

The required band is the interval between the survey-anchored back-cast and the all-in option bound (Section 6). For 2026 it is the envelope of Section 5, whose central estimate is 15% (90% interval 12–18%).

Three readings of this table, in increasing order of importance.

Why the bands are wide early. The honest uncertainty in 2013–2017 is *who* the marginal holder was. Under the survey-anchored back-cast, participants were few but each held a modest weight, and the required return was only 10–15%: on that reading the wedge was enormous, 100-plus points a year; the market simply did not anticipate the adoption that came, and the realized returns were a surprise, not a payment. (In the language of the companion scarcity framework (Papadogiannis 2025), where network value is desirability times scarcity amplified by adoption, the wedge is the unanticipated growth of that value function; this paper prices the risk side, that framework describes the value side.) Under the all-in reading, which the 2017–22 option bounds support for the segmented era, required returns were themselves in the tens-to-hundreds, and the wedge was modest. The truth sits between, and for the paper's purpose it does not need to be settled: on *either* reading the early engine (concentration, volatility, segmentation, surprise) has measurably shut down.

Which term carried the requirement, and when. Figure 6 takes the midpoint of the band and splits it into the four pieces of equation (1), in levels and as shares. The picture is lopsided in a way the tables cannot show. In 2013 the concentration term, $\gamma\sigma^2 w$, was about 95% of a requirement near 125% a year; the risk-free rate, the beta term, and the mortality term together were a rounding error. Concentration still supplied four-fifths of the requirement in 2021. Only after the ETF-era dilution of the holder base and the

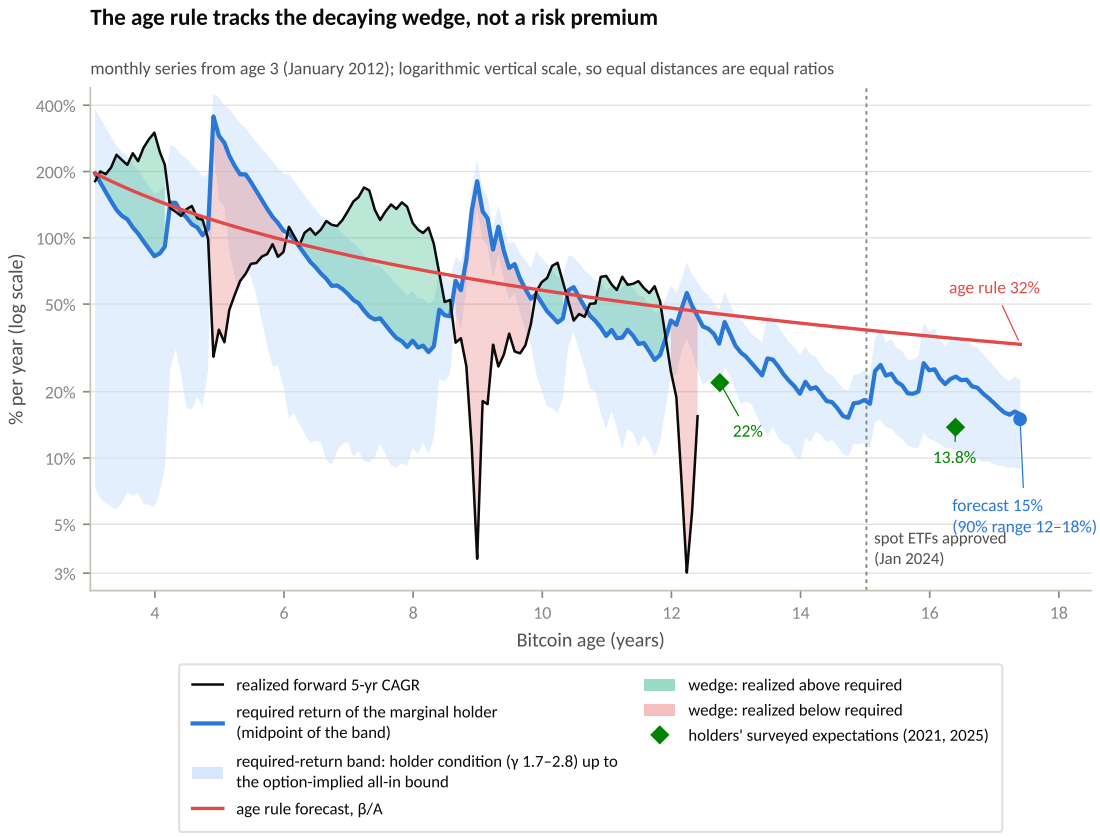


Figure 5: Required return versus realized forward five-year drift versus the age rule, on a logarithmic vertical scale so that equal distances are equal ratios. Series begin at age 3 (January 2012), the start of the companion paper's backtest sample and the first date with enough price history for a trailing volatility estimate; the realized curve ends at age 12.4 because each of its points is the compound return over the following five years. The shaded gap between realized and required is the wedge: positive early, negative after late 2021. Green diamonds mark holders' surveyed expectations; the dotted line marks spot-ETF approval (January 2024), Bitcoin's market-integration event; the right edge marks the 15% forecast against the age rule's 32%.

fall in volatility did the floor terms become visible: in 2026 the risk-free rate is about 30% of the midpoint requirement, the beta term 17%, mortality 10%, and concentration 43%. Two things follow. First, the history of Bitcoin's required return is almost entirely the history of one term, which is why Section 2 could build the equation around it. Second, the term that has not fallen, mortality, is the one that now matters most relative to its size: a fixed 0.3–3 points was invisible against 125% and is a tenth of 15%. (The midpoint path ends at 16% rather than the 15% central estimate of Section 5 because the midpoint weights the band's corners equally, which the propagation of Appendix B.5 does not.)

Why the age rule backtested so well, and why it has started to miss. Both the required return and the wedge decayed over 2012–2021, roughly together, and a one-parameter power law in age is an excellent smoother of a sum of decaying curves. That is the age rule: a reduced form of required-plus-wedge, not a structure. The reduced form has no engine of its own, so when the components stopped decaying *together*, it started to miss, and in the one direction the decomposition predicts. From late 2021, holders required 21–58% (their own first-order condition; the option bound was higher still) and received about 13%: from

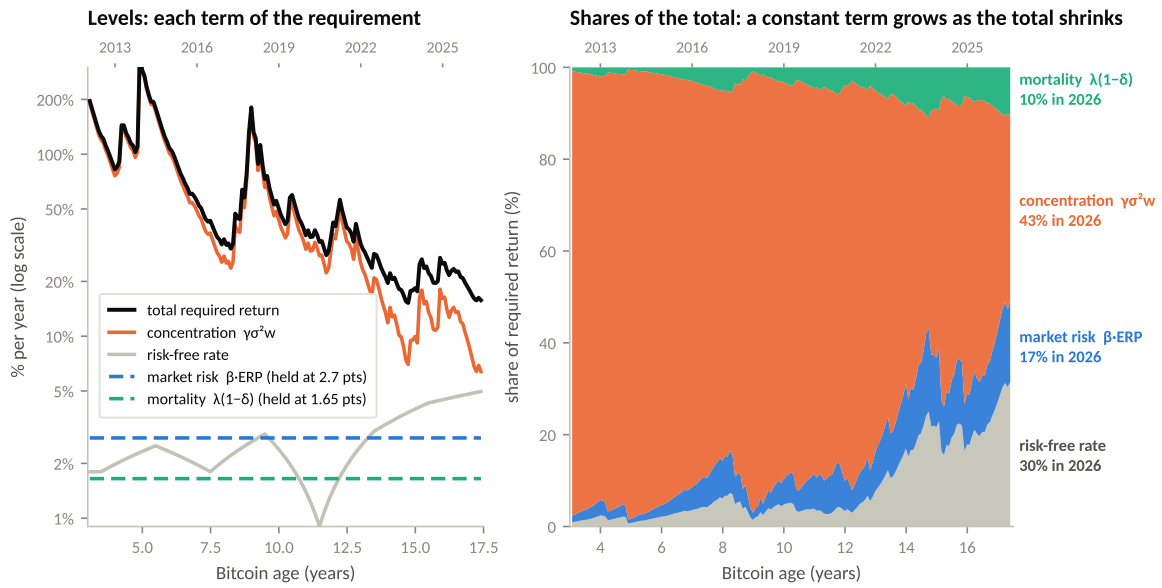


Figure 6: The four terms of equation (1) over Bitcoin's life, from the midpoint of the required-return band. Left: the level of each term and of their sum, in percent a year on a logarithmic scale. Right: each term's share of the total; because the total falls from about 120% to 17%, the shares of the two constant terms rise even though the terms themselves do not move. The beta and mortality terms (dashed) are held constant at the midpoints of their stated ranges, 2.7 and 1.65 points, because neither has a defensible time path: beta estimates depend on the window and the segmented-era beta is not what a holder faced, and the elite hazard is one average over 2014–2026 resting on three events. Concentration is the remainder, which includes the all-in option bound at the band's upper edge.

the 2021Q3 survey date to May 2026 Bitcoin compounded at 12.8% a year, and from its November 2021 peak at 4%. **The wedge turned negative.** The companion backtest paper's own out-of-sample record shows precisely this: in windows beginning after the power law's 2019 publication, Bitcoin fell short of the age rule by 5–8 points a year. What looked there like a small, possibly transient overshoot is, in this decomposition, the sign flip of the wedge.

Why the wedge is unlikely to be a hidden risk premium. Three facts narrow that escape route. The graveyard caps the mortality channel at ∼0.3–3 points (Section 4) and rejects the declining-hazard shape outright. The $\sigma^2 w$ concentration channel is measured, and today contributes 2–9 points. And, in a test we ran against ourselves, the concentration quantity has *no forecasting power of the right sign*: regressing forward 3–5 year returns on real-time $\sigma^2 w$ alongside the age variable, $\sigma^2 w$ enters negative (market cap sits in w's numerator, so the product spikes at mania tops) while the age term is undisturbed. The marginal-holder condition is an equilibrium restriction holding at each date, not a timing signal, and the paper does not claim it to be one. What these facts cannot exclude are premia for risks outside both measured channels, such as ambiguity aversion or disaster states never realized in 2010–2026; if present, they must show up where all priced tail risk shows up, in option prices, and Section 7 sends the reader there. Full regressions in Appendix B.

Risk premium or exuberance? What the wedge can tell apart. The concentration term is mechanical:

a holder with a large fraction of her wealth in one volatile asset bears a great deal of undiversified risk, and $\gamma\sigma^2 w$ is the return that makes keeping it rational, whatever put her there. Two different stories produce the same w. In Merton's (1987) story only a few investors can or will hold the asset, so each must hold a lot of it and the price must offer them a premium for the risk they are stuck with; $\gamma\sigma^2 w$ is then a risk premium in the proper sense, and it is paid on average. In the other story investors pile in because they expect a great deal, so their weights are the sizing rule of Section 9 run on inflated beliefs, $w = (\hat{\mu} - r_f)/(\gamma\sigma^2)$; $\gamma\sigma^2 w$ then recovers what they expected rather than what the risk was worth, and because the price has already risen to meet the belief, the return does not arrive on average. From the weight alone the two are indistinguishable, which is why the consistency check of Table 3 shows coherence between what holders held and what they said, not that either was right. What separates them is what happened next, and that is what the wedge records. Where the premium is real, realized returns match or exceed it; where it was exuberance, they fall short. On this reading the early years cannot have been exuberance, because realized returns exceeded even the all-in option bound, which no optimism about w can produce; they were surprise. Late 2021 reads the other way: positions implied 21–58%, holders said 22%, and 13% arrived, so part of the 2021 "requirement" was belief dressed as premium, and the negative wedge is that belief being corrected. The general statement, which is the paper's distinction with a cause attached, is that exuberance raises the holder's position, the position raises the subjective hurdle, and the price rises with it, which lowers the objective expected return. It also gives the equilibrium hypothesis of Section 7 a sharper content: that today's weights of 8–19%, diluted by the ETFs toward investors holding a sliver, are participation-driven rather than belief-driven, so that today's 15% is a premium that will be paid and not a hope that will not. The option-implied bound is the independent witness here, since risk-neutral variance does not depend on holders' optimism, and it sits at 16–30% for an all-in holder, above the 15%.

The cross-section tells the same story from the side. If early crypto returns had been mostly required premium, the typical altcoin (more volatile, more concentrated, higher λ than Bitcoin) should have delivered even more. It delivered less than nothing: across the significant-asset cohorts, the *median* coin born 2015–2020 shows a negative compound return at every horizon, while cohort *means* stay large on the back of a few winners, exactly the lottery shape that a 10–17% annual death rate plus a handful of extreme survivors produces. Within Bitcoin's own 2010–14 birth cohort at age ten, the mean coin compounded at about +167% a year, the median at +50%, and Bitcoin at +233%. Conditioning on the winner, which every rule fitted to Bitcoin's history silently does, is worth on the order of 180 points a year over the in-cohort median at that age. The wedge, in other words, is visible twice: over time in Bitcoin's own decomposition, and across assets as the gap between the winner and the median.

7. THE FORECAST, AND WHAT WOULD PROVE IT WRONG

A required return is a statement about risk; a forecast is a statement about the price process. The bridge from one to the other is an equilibrium condition, which we state as our hypothesis and own: *Bitcoin is now priced in equilibrium by its marginal holder*, with no systematic underpricing left in it. The expected return of any asset is its market required return plus whatever predictable drift mispricing adds; surprises, by definition, average to zero in advance. The historical wedge of Section 6 mixed three things that one price path cannot separate: adoption news the market had not priced, the selection of one survivor from many, and possibly a period of genuine underpricing while the asset was segmented from mainstream capital. The first

two cannot be expected to recur; the hypothesis is that the third has closed, and that today's holder weights reflect participation rather than exuberance (Section 6), which is what the ETF era, the dilution of the holder base, and the compression of the option bounds all indicate. Under that hypothesis, and only under it, the expected return equals the required return of the marginal-holder condition, every ingredient of which is now observable:

If Bitcoin is priced in equilibrium, its expected return for September 2026 to September 2031 is about 15% a year, with a 90% range of 12–18% (full envelope 9–22%). The age rule says 32% (mean-type), 22–28% compounded. The two forecasts differ enough to be separated by a single five-year window, and we commit to ours in print.

To keep the claim honest, here is what each outcome would mean:

- **BTC compounds (CAGR) at $\geq\sim$25% a year to 2031.** The age rule is vindicated and this paper's decomposition is wrong in an instructive way: either a required-return component we measure as small is in fact large (which the options market would have to show; watch whether implied variance and holder concentration stop compressing), or Bitcoin remains systematically underpriced, so that a predictable drift above the required return persists, which is the age rule's implicit claim.
- **BTC compounds in the single digits to low twenties.** The decomposition stands; the age rule's first structural break is confirmed, in the direction and for the reason stated here.
- **BTC compounds near zero or below.** Within our band's logic this is a bad draw of the (large) diffusive risk, not a refutation. But if it comes with *rising* option-implied crash intensity well above the graveyard's 0.3–3%, the mortality term was underestimated.

Two notes on the test itself. *Units:* our 15% (90% range 12–18%) is an arithmetic drift; at today's $\sigma \approx 42$–45% the volatility drag is roughly $\sigma^2/2 \approx 9$ points, so the compounded (CAGR) counterpart is about 6%, with a 90% range of 3–9% and an envelope of 0–13%, against the age rule's compounded 22–28%. In matching units the two forecasts do not overlap, and a single five-year window grades them. *Asymmetry:* as the three outcomes above make explicit, the forecast is sharply falsifiable upward but only conditionally falsifiable downward. A one-and-a-half-sided test is the price of forecasting a 45%-volatility asset, and we state it rather than hide it.

What about the halving cycle? Practitioners will ask where Bitcoin's four-year "halving cycle," the pattern of peaks in 2013, 2017, and 2021 that followed the supply halvings of 2012, 2016, and 2020, fits in this account. Four points. First, the halving changes supply, and supply does not appear in equation (1): cutting issuance from 0.8% to 0.4% of the outstanding stock a year alters none of the risks a holder is paid for, so the required return is unaffected by the schedule; supply matters for the price level, which this paper does not model. Second, the schedule is written in the code and known years ahead, and in an equilibrium a known future supply change is in today's price and cannot carry a predictable excess return around its date. A reliable, tradeable cycle would therefore be a systematic mispricing, and the equilibrium hypothesis above says there is none left; the forecast is, among other things, a bet that the cycle is not a tradeable regularity going forward. Third, the past cycles were real and are in Figure 5: the five-year windows starting at the 2013, 2017, and 2021 peaks are the troughs of the realized curve and the red wedges, those starting in the 2015 and 2019 troughs are the green ones. In this paper's language the cycle is the wedge alternating sign, surprise or exuberance followed by correction, and the condition says nothing about its timing, only

that its mean is zero in equilibrium; the regressions of Appendix B.2, where the concentration quantity spikes at mania tops and predicts low forward returns, are the bust leg of that cycle read as exuberance correcting rather than as a halving clock. Fourth, the paper predicts that the cycle's amplitude and predictive content should fade, because the mechanism that produced it, a small concentrated holder base at 80–150% volatility, is the mechanism documented compressing here and in Figure 7, and because an integrated market arbitrages a four-year calendar. The forecast window, September 2026 to 2031, straddles the 2028 halving and whatever peak or trough follows it, so the cycle phase is partly averaged out of the test, which is one more reason for the five-year horizon.

Two auxiliary predictions sharpen the mechanism. First, drift compression should be fastest where volatility and holder concentration fall fastest: Ether post-ETF should show the same compression; a still-segmented, still-concentrated large asset should not. Figure 7 shows the first check already passes: Ether's volatility and forward returns compress alongside Bitcoin's into the ETF era. Second, option-implied crash intensity should remain in the low single digits, consistent with the measured elite jump hazard; a reading persistently above $\sim$10% a year would push our band up and signal that the market disagrees with the graveyard, whether about contagion, regulation, or forward-looking tail risks such as a quantum break of Bitcoin's signature scheme (Appendix C.3), which realized mortality cannot yet contain.

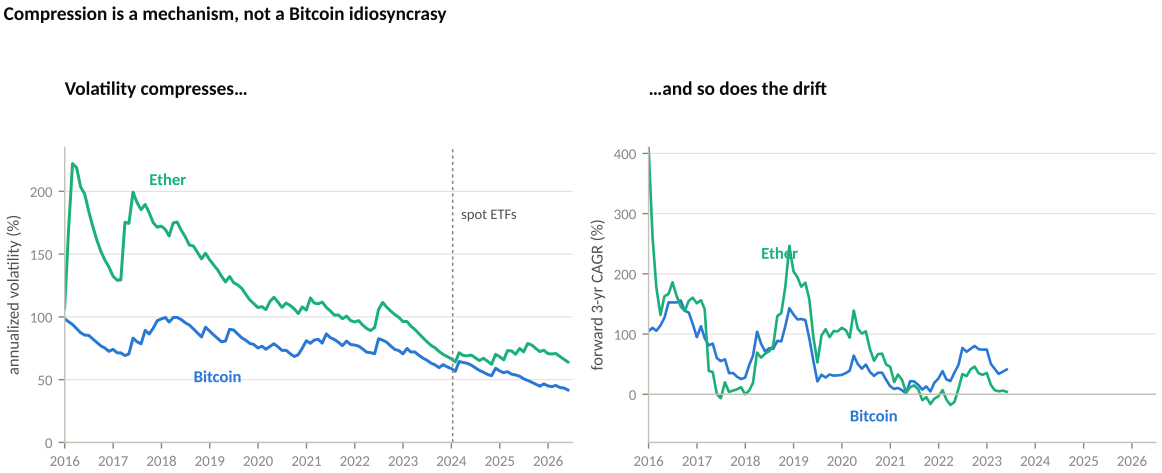


Figure 7: The compression mechanism is not Bitcoin-specific. Left: EWMA annualized volatility for Bitcoin and Ether, with spot-ETF approval dates marked. Right: realized forward three-year returns for both assets.

8. THE SAME PRICING FOR ETHER AND OTHER CRYPTOASSETS

Nothing in equation (1) is specific to Bitcoin. The condition prices any asset held by a concentrated investor who faces a risk of total loss; what changes from one cryptoasset to the next is the value of the inputs. The graveyard was built from the whole asset class, so it already supplies the death-risk input by size class, and Bitcoin's volatility and holder-weight measurements have natural counterparts. Table 5 walks the condition down the size ladder.

Ether. Ether has been a top-five asset for a decade, so its death risk sits in the same elite class as Bitcoin's and the mortality term is the same 0.3–3 points. Its volatility is higher, 64% against 42% today, and that is

Table 5: The marginal-holder condition across the cryptoasset size ladder. Required returns per year. The central estimate and 90% interval come from drawing each input independently and uniformly over its stated range (Appendix B.5); the envelope sums the extremes. Risk-free 5% and beta 0.33–1.05 on a 4% equity premium throughout; $\gamma \in [1.7, 2.8]$. Altcoin holder weights are illustrative, not surveyed; small-coin volatility is taken as 100–130%.

	Bitcoin	Ether	Large altcoin	Small coin
Volatility σ, 2026	42%	64%	∼80%	100–130%
Holder weight w	0.08–0.19	0.08–0.19	0.10–0.30	0.10–0.30
Death hazard, per year	0.3% jump	0.3% jump	3.2% all-cause	17.4% all-cause
Mortality term	0.3–3 pts	0.3–3 pts	1–3 pts	16–17 pts
Required return, central	**15%**	**22%**	**38%**	**82%**
90% interval	12–18%	16–28%	25–55%	54–124%
envelope	9–22%	12–34%	18–66%	40–165%
Delivered	∼13%	$\sim -7\%$	-7%	not observable

Delivered: Bitcoin and Ether, compound annual return from the 2021Q3 survey date to May 2026 (Section 6); large altcoins, the median compound return of the significant-asset cohorts born 2015–2020, measured five to seven years after birth (-6.5% to -8.7% a year, Section 6). Small coins have no usable price history in the data; what is observable is that half of the smallest quartile is dead within 3.3 years (Figure 3). Hazards are the graveyard's size-quartile and elite estimates (Section 4).

the whole difference. On the holder weight we make one assumption: surveys report a household's crypto holdings as a block, Ether's monthly returns have a correlation with Bitcoin's of about 0.8, and so the share that matters for a holder's risk is her crypto share, the same 8–19%. The concentration term then scales with the ratio of variances, $(0.64/0.42)^2 \approx 2.3$, and Ether's required return comes out at about 22% a year (90% interval 16–28%; envelope 12–34%). Over the same window used for Bitcoin, from the 2021Q3 survey date to May 2026, Ether delivered about -7% a year (the endpoint matters at this volatility: measured over the three years from mid-2021 to mid-2024 it is $+15\%$), far below the central estimate on either reading; in the language of Section 6, its wedge, like Bitcoin's, has been negative since 2021. Figure 7 showed Ether's volatility and realized drift compressing alongside Bitcoin's; the condition says the required return has compressed with them.

Large altcoins. For the surviving assets in the largest size quartile outside the top two, volatility is about 80% and the all-cause death rate is 3.2% a year, mostly slow fades in which holders recover part of their value, so a mortality term of 1–3 points. The holder weight is not surveyed for individual coins; concentration among committed communities is plausibly higher than for Bitcoin, and we show an illustrative 0.10–0.30. The required return lands at about 38% a year (90% interval 25–55%; envelope 18–66%). The significant-asset cohorts of Section 6, which are the surviving large altcoins, delivered median compound returns of -7% to -9% a year five to seven years after birth: the typical large altcoin has not paid what its risk required, by some 45 points a year.

Small coins. At the bottom of the ladder the mortality term stops being a footnote. The smallest quartile dies at 17.4% a year, with recovery close to nothing, so the death risk alone adds about 16 points to the required return before any compensation for volatility of 100% or more. The condition demands about 82% a year (90% interval 54–124%), and 40% at the very least. What was delivered is not observable in the price data, which the free tier does not carry for small coins; what is observable is that half of them are dead within 3.3 years of listing (Figure 3), and that even the surviving significant altcoins, a far stronger group, lost money at every horizon. Holders of small coins were systematically not compensated for the risks the condition prices. The pattern is the one documented for lottery-like stocks, where investors accept low

average returns for a small chance of an extreme payoff (Barberis and Huang 2008); the graveyard prices that preference in this asset class: for a small coin the gap between required and delivered return is at least the 45 points measured for the surviving large altcoins, and for the half that die within three years it is the whole stake.

The altcoin holder weights are illustrative and the realized comparisons cover only surviving assets with price history, which flatters them (Section 10); the direction of the result does not depend on either. As one moves down the size ladder the required return rises steeply, because volatility and death risk both rise, while realized returns fall. Bitcoin is the one cryptoasset for which the two sides of the condition have been close, which is a statement about Bitcoin's holder base and survival record, not about the condition.

9. FOR THE ALLOCATOR

The decomposition translates directly into practice. This section states the hurdle a diversified allocator should apply, inverts the condition into a sizing rule, places the result in a Black–Litterman frame, and then asks the question the numbers invite: whether, and for whom, Bitcoin is worth holding at all, against equities and against gold.

The hurdle. A diversified allocator should require of Bitcoin what its diversifiable-risk-adjusted contribution costs her: the risk-free rate plus beta times the equity premium, about **7–9% today**, plus some or all of the mortality term, 0.3–3 points, to the degree she judges crypto mortality systematic rather than diversifiable. The 2022 clustering of elite deaths (Terra, Celsius, FTX within seven months) argues it is partly systematic; we would charge 1–2 points. Call the hurdle 8–11%. This is not a forecast of what Bitcoin will do; it is the bar it must clear to belong in her portfolio at more than market weight.

The sizing rule. The marginal-holder condition, inverted, sizes the position. If an investor's own view of Bitcoin's expected return is $\hat{\mu}$, her risk aversion γ, and Bitcoin's volatility σ, her weight should be

$$w^* = \frac{\hat{\mu} - r_f - \beta\,\mathrm{ERP} - \lambda(1-\delta)}{\gamma\sigma^2}. \tag{3}$$

Worked example: an investor with $\gamma = 3$ who believes our central forecast, $\hat{\mu} = 15\%$, at today's $\sigma = 42\%$ ($\sigma^2 \approx 0.18$): $w^* = (15 - 5 - 2.7 - 0.3)/(3 \times 0.18) \approx$ **13% of financial wealth**. That is aggressive by institutional standards, and about two-thirds of the 19% the average 2021 holder actually ran at far higher volatility. An investor who believes the age rule's 32% gets $w^* \approx 45\%$; the distance between those two positions is the practical cost of getting this paper's question wrong.

The Black–Litterman translation. In a Black–Litterman portfolio, the tilt away from market weight is proportional to the gap between your expected return and the market-implied one (Black and Litterman 1992). This paper reduces that gap to observable disagreements: your volatility path, your view of holder-base integration (the path of w), and your mortality estimate against the market's option-implied one. An allocator with no disagreement on those three has no basis for holding Bitcoin away from its market weight, currently well under 1% of global financial wealth, however glorious its history.

Is it worth holding at all? The results invite the question, because broad equity indices offer a similar reward per unit of risk without any risk of dying, and for the smaller cryptoassets the answer is plainly no: a required return of 38–82% against delivered returns that are negative or unobservable means their holders have been systematically underpaid, and the only coherent account of their demand is the lottery preference

of Section 8. For Bitcoin the answer depends on who is asking. The 15% is what its concentrated holders require, and in equilibrium it is what they get, so for them it carries no alpha; "fairly priced" means exactly that. A diversified investor is in a different position. She does not bear the concentration risk those holders are paid for, and at a market weight near 1% she barely bears the death risk either: a jump to zero would cost her 1% of wealth, uncorrelated with anything else she owns. Her hurdle is the normative 8–11% above, and the asset offers about 15%. The 4–7 point difference is a rent, the premium a segmented holder base demands, available to anyone able to hold the asset as a small part of a diversified portfolio; that is Merton's (1987) original observation about limited participation, and it is why the sizing rule above returns a weight of about 13% for a $\gamma = 3$ investor who accepts the central forecast, rather than zero.

Against equities, standing alone. Standing alone, Bitcoin is not a better bet than equities, and the comparison can be made with the paper's own inputs. Take the S&P 500, the index against which Bitcoin's beta is measured. Its expected excess return is the equity premium used throughout, 4% (6% in the high case); its volatility is 16–20% a year over the long historical record (Dimson, Marsh, and Staunton 2002) and 16.5% in the current J.P. Morgan assumptions (J.P. Morgan Asset Management 2026). The ratio of excess return to volatility, the Sharpe ratio (Sharpe 1994), is therefore about 0.20–0.25 (4% over 16–20%) and up to about 0.35 in the high case. Bitcoin at a 15% arithmetic required return, a 5% risk-free rate, and 42% volatility gives $(15-5)/42 \approx 0.24$, and about 0.22 once the mortality term, which is compensation for a loss and not a return, is taken out. The two sit within rounding of each other, and the choice of index does not change the conclusion: a growth index such as the Nasdaq 100 carries a similar premium at higher volatility, and a global all-cap index (FTSE Global All Cap or MSCI ACWI IMI, some 10,000 stocks across developed and emerging markets) nearly the same figures, since the J.P. Morgan assumptions give world equity an arithmetic return of 8.3% at 16.8% volatility, a ratio of about 0.20 over the 5% risk-free rate, against 0.18 for US large cap (7.9% at 16.5%) on the same basis. A reader who prefers the historical US premium of 7–8 points over bills (the familiar 10% nominal return of the last century, which includes a dividend yield of 4% rather than today's 1%, higher inflation, and a tripling of valuation multiples) should raise both sides: equities' ratio rises to about 0.45, and Bitcoin's beta term rises with the same premium, taking its required return to 17–19% and its ratio to about 0.30. On the historical premium equities win the standalone comparison; on the forward-looking premium used here they tie; neither changes the allocator's case, which rests on correlation. What Bitcoin adds to a portfolio is a correlation with equities below 0.2 (Liu and Tsyvinski 2021), and that is the entire source of its value as a holding; if integration eventually raises the correlation toward equity-like levels, the case shrinks with it. What this paper removes is not the case for a small Bitcoin position but the case built on 32%: the belief that Bitcoin offers an equity-like reward-to-risk ratio at triple the return. At a central 15% arithmetic, about 6% compounded, it is a reasonable diversifier and nothing more.

In Sharpe units. Dividing equation (1) by σ turns the required return into a required reward per unit of risk, and because $\beta = \rho\,\sigma/\sigma_m$ the market term simplifies:

$$\frac{\mu - r_f}{\sigma} \;=\; \rho\,\mathrm{SR}_m \;+\; \gamma\sigma\,w \;+\; \frac{\lambda(1-\delta)}{\sigma}, \tag{4}$$

where ρ is the asset's correlation with the market and SR_m the market's Sharpe ratio. For a fully diversified holder of an asset that cannot die the right side collapses to $\rho\,\mathrm{SR}_m$, the CAPM in Sharpe units: no asset can offer more reward per unit of risk than the market times its correlation with it. The concentration term,

$\gamma\sigma w$, is what a holder adds when she owns a lot; σw is the position's contribution to her portfolio volatility, so Sharpe stops being scale-free. The mortality term shrinks with volatility, because a fixed expected loss is a smaller share of a larger risk budget. More generally, at any holder's optimum an asset's Sharpe ratio equals its correlation with her own portfolio times that portfolio's Sharpe ratio; concentration is simply what makes the correlation high. The numbers are telling. Today, with $\rho \approx 0.2$ and $\mathrm{SR}_m \approx 0.25$, the correlation term is about 0.05, the concentration term ($2.25 \times 0.42 \times 0.13$) about 0.13, and mortality (1.65 points over 42%) about 0.04: a required Sharpe ratio of about 0.22, the figure reached above by division. In 2013 the band midpoint implies about 0.8, and above 1 at its all-in edge, a reward per unit of risk that no asset class delivers for long and Bitcoin did not. Once holders are diluted toward market weight ($w \to 0$) the requirement falls to $\rho\,\mathrm{SR}_m$ plus the mortality term, about 0.09 at today's volatility: an asset 0.2-correlated with equities should, in an integrated market, earn about a fifth of the market's reward per unit of risk plus a few points for death risk. Bitcoin's current Sharpe ratio matches equities only because its holders are still concentrated, and integration should push it below equities, not toward them. The same form answers the allocator's question directly: an asset improves a diversified portfolio's Sharpe ratio if and only if its own Sharpe ratio exceeds ρ times the portfolio's, here 0.22 against 0.05, so a small position adds today, and the gain is exactly the limited-participation rent, which vanishes as $w \to 0$ by construction. For gold, with $\rho \approx 0$ and a Sharpe ratio of a few hundredths, the improvement is negligible on expected-return grounds; whatever case gold has rests on tail behavior this condition does not price.

Appendix C applies the same condition to gold, where it reproduces a known result (a required return near the risk-free rate; Barro and Misra 2016), and spells out what "digital gold" would mean in return terms: about 10% a year at gold's volatility and holder dilution, and gold's 5–6% only if Bitcoin also shed its beta and its death risk.

10. LIMITATIONS AND THE MEASUREMENT AGENDA

This is a measurement paper, so its limits are measurement limits; we list them in the order a critical reader should weigh them.

1. **The holder weight before 2021 is bounded, not measured.** The portfolio-share surveys begin in 2021; everything earlier rests on market-clearing back-casts and option bounds, and the 2026 lower bound on an assumed weight for ETF and treasury investors (Appendix D). The paper's historical bands are honest about this width, and its forward claim does not depend on the early years. The single most valuable upgrade would be a measured w path: entity-adjusted on-chain concentration joined to holder-wealth estimates, the 2025 portfolio-share figures in the Candia et al. online appendix (not yet published in text at this writing), and the ETF share of total holdings.

2. **Coverage-onset selection in the graveyard.** Assets enter the panel only after becoming significant, so young-age hazards are understated. This biases the data *toward* Lindy, which makes the rejection conservative, and it leaves the elite estimates untouched; but population hazard *levels* at ages under two should be read as floors.

3. **Death timing and classification.** Feed and volume deaths lag economic death by months; four elite "deaths" are contested zombies (chains that operate far below their manic highs); one (ICP) is an

artifact of a listing-day price spike. All classifications, dates, recovery fractions, and confidence flags are in Appendix A, and the headline elite hazard moves only from 3.5% to 3.15% when the contested cases are dropped.

4. **Bitcoin's own existential era is unmeasurable.** No cross-section existed in 2010–2014 to price the risk that Bitcoin itself was a dead end. Our early-period required-return bands are therefore wide by construction, and nothing in the paper's forward claim rests on resolving them.

5. **Survey-to-wealth pairing.** The participation series is US-based and the wealth aggregates global; re-running the clearing calculations with US household wealth against US participation moves the bands by amounts well inside their width (Appendix B).

6. **The holder weight is endogenous to beliefs.** A weight chosen on over-optimistic expectations satisfies the same first-order condition as one forced by segmentation, so the concentration term recovers the holder's subjective hurdle; whether that hurdle was a fair premium is adjudicated by realized returns (the wedge of Section 6) and by the option-implied bound, not by the equation. The forecast assumes that the ETF-era weights are participation-driven; that assumption is what the 2031 test checks.

7. **Few independent windows.** Five-year returns from a 16-year price history contain roughly three independent observations. That is why the paper leans on levels at measured dates, cross-sectional mortality, and two survey waves rather than on time-series regressions, and why its forecast is phrased so that a single new window can grade it.

8. **Illustrative inputs for the other assets.** The altcoin holder weights of Section 8 and gold's holder weight in Appendix C are assumed ranges, not survey measurements, and gold's mortality allowance is a judgment; the required returns built on them are directional, and the Bitcoin estimate, which rests on measured inputs, does not depend on them.

11. CONCLUSION

The riddle was that Bitcoin seemed to carry three expected returns at once: 32% from its own tested history, 6–18% from the CAPM, 14–22% from its holders' mouths. The resolution begins with classification. Only the first was ever an expected return, a property of the price process. The CAPM range is a normative required return, the hurdle of an investor who does not hold Bitcoin. The holders' numbers are the market's required return, the compensation demanded by the investor who does, and they move, wave by wave, in the direction and magnitude the marginal-holder condition, Merton's (1987) limited-participation pricing with measured inputs, says they should: with her volatility and her portfolio weight, at ordinary levels of risk aversion. And the 32% is the smooth curve through a history equal to that required return plus a wedge of unanticipated adoption and winner-selection. The graveyard shows the wedge was never mortality compensation (hazard rises with age; the top-20 jump hazard is 0.3% a year); the cross-section shows it was never available to the typical asset (the median coin lost money); and Bitcoin's own record shows it turned negative after 2021.

What remains, if the asset is now priced in equilibrium, is a required return built from observable things: 40-some percent volatility, a holder base diluting toward market weight, a small and datable mortality risk, and a risk-free rate. Those ingredients say about 15% a year for 2026–2031, with a 90% range of 12–18%; the same condition, fed gold's inputs (Appendix C), returns the risk-free rate plus half a point, which is what "digital gold" would mean if Bitcoin ever got there. If Bitcoin instead repeats its age-rule 32%, then either it was still underpriced, or a risk we measured as small was large, and the framework here says exactly which observables separate the two. Either way, the question that opened the companion paper (compensation for risks that never came, or an asset the market underpriced?) now has an evidence-based answer: on either reading of the early marginal holder, mostly the second, for a decade; neither, anymore.

For practitioners. Four takeaways. Use 15% a year (12–18%) as Bitcoin's expected return in capital-market assumptions, not the 32% its history extrapolates, and about 6% in compound terms; at 42% volatility this is an equity-like reward per unit of risk, not a superior one. Treat the 4–7 point gap between that figure and a diversified investor's 8–11% hurdle as a limited-participation premium that is available to small, diversified positions and will shrink as ETFs broaden the holder base. Size the position with the condition inverted (equation 3): about 13% of financial wealth for an investor with risk aversion of 3 who accepts the central forecast, and proportionally less for higher risk aversion or a lower view. And charge for mortality explicitly, 0.3–3 points, with option-implied crash intensity as the running check on whether the market has begun to price a larger hazard. The measurement agenda of Section 10, above all a measured path for the holder weight, is where the estimate can be sharpened; the forecast itself is graded in September 2031.

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DECLARATION OF GENERATIVE AI USE

Generative AI (Anthropic's Claude) was used under the author's direction to write and test the code in the replication package, to produce the figures from that code, and to draft and edit text. The research design, the curation of the death classifications, all interpretations and conclusions, and the final text are the author's, who has verified the outputs and takes full responsibility for the content. Generative AI was not used to generate or alter data.

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Data and code availability: the full replication package (panel-construction and survival scripts, the curated asset-level panel, all result tables, and every exhibit) is maintained at `https://github.com/apapadogiannis/bitcoin-required-return`. The pipeline ends with an audit script that recomputes and checks every quantitative claim in this paper. Underlying market data are the public Coin Metrics community dataset (snapshot ending May 24, 2026), which the package downloads from its source rather than redistributing.

A. THE CURATED ELITE DEATH TABLE

All 36 deaths among the 219 ever-top-20 assets, with the classification each received. *Type*: jump = sudden, value-destroying collapse priced as a Poisson event; exchange = failure of the issuing exchange; fade = gradual abandonment and loss of listings; contested = flagged by a mechanical rule but the chain still operates. δ is the approximate fraction of pre-event value recoverable by a holder (last liquid price or successor claim as a share of pre-event value); it is an estimate, not an accounting figure. *Conf.* flags classification confidence (H/M/L). Dates are the curated economic death dates; mechanical rules alone would misdate LUNA, UST, and FTT, whose collapses are entered from the public record.

Asset	Death date	Type	δ	Note	Conf.
LTL	2014-12	fade	$\sim$0	minor 2013-era coin	M
FCT (Factom)	2021-07	fade	$\sim$0.05	sponsor company bankrupt 2020; chain stalled	H
LBC (LBRY)	2021-09	fade/regulatory	$\sim$0.05	SEC action 2021; project wound down	H
PPT (Populous)	2021-11	fade	$\sim$0	abandoned	M
ETP (Metaverse)	2022-01	fade	$\sim$0	abandoned	M
NCASH (Nucleus Vision)	2022-01	fade	$\sim$0	delisted majors 2021	H
ZB	2022-02	exchange	$\sim$0	ZB.com halted withdrawals 2022	H
WTC (Waltonchain)	2022-04	fade	$\sim$0	abandoned	H
UST (TerraUSD)	**2022-05-12**	**jump**	$\sim$0.02	algorithmic stablecoin depeg	H
LUNA (Terra)	**2022-05-12**	**jump**	$\sim$0.01	supply hyperinflation; LUNA2 airdrop gave partial recovery	H
QASH (Liquid)	2022-07	exchange	$\sim$0	exchange hacked 2021, absorbed into FTX	H
QC (Qcash)	2022-08	fade	$\sim$0	abandoned	M
FTT (FTX Token)	**2022-11-10**	**jump**	$\sim$0.05	FTX insolvency	H
MITH (Mithril)	2023-03	fade	$\sim$0	serial delistings	H

Asset	Death date	Type	δ	Note	Conf.
QSP (Quantstamp)	2023-05	fade	$\sim$0.02	token abandoned	M
SEELE	2023-07	fade	$\sim$0	abandoned	M
VTC (Vertcoin)	2023-08	fade	$\sim$0.02	lost major listings	M
MDA (Moeda)	2023-09	fade	$\sim$0	abandoned	M
BTS (BitShares)	2024-06	fade	$\sim$0.02	community remnant only	M
AE (Aeternity)	2024-10	fade	$\sim$0.02	team wind-down	M
BCN (Bytecoin)	2024-10	fade	$\sim$0	long dormant; delisted majors years earlier	H
OMG (OMG Network)	2025-02	fade	$\sim$0.02	Binance delisting; development ceased	H
KEY (SelfKey)	2025-04	fade	$\sim$0	abandoned	M
LOOM (Loom Network)	2025-05	fade	$\sim$0	abandoned	H
XEM (NEM)	2025-05	fade	$\sim$0.05	major delistings; earlier opt-in XYM airdrop (2021)	H
BTG (Bitcoin Gold)	2025-06	fade	$\sim$0.02	Binance delisting	H
LINA (Linear)	2025-06	fade	$\sim$0	project shut down	M
PDA (PlayDapp)	2025-06	fade	$\sim$0.02	post-2024-exploit decline (PLA$\rightarrow$PDA swap preceded)	M
YFII (DFI.Money)	2025-07	fade	$\sim$0	abandoned fork	M
FLOW	2025-09	**contested**	n/a	chain operating; >99% below ATH	H
MXC	2025-09	fade	$\sim$0.02	rebranding/decline	L
STMX (StormX)	2025-10	fade	$\sim$0	abandoned	M
NANO	2025-12	**contested**	n/a	active community; lost major listings	M
ICP (Internet Computer)	2026-01	**contested**	n/a	alive, top-100; "death" is an artifact of the listing-day price-spike ATH	H
HIFI	2026-01	fade	$\sim$0	delisted	L
SUSHI (SushiSwap)	2026-03	**contested**	n/a	protocol operating; >99% below ATH	H

Headline sensitivities. All 36 deaths: hazard 3.54%/yr (CI 2.48–4.90). Excluding the four contested cases: 3.15%/yr (CI 2.15–4.44). Jump-collapses only (LUNA, UST, FTT): 0.30%/yr (CI 0.06–0.86). In the ever-top-50 universe the jump set adds SRM (Serum) and ANC (Anchor), both 2022 casualties of the same two cascades, for 0.23%/yr.

Migrations excluded as deaths (holders received successor value): MATIC$\rightarrow$POL, EOS$\rightarrow$Vaulta (A), FTM$\rightarrow$S (Sonic), GNT$\rightarrow$GLM, RNDR$\rightarrow$RENDER, BTT redenomination, NU$\rightarrow$T (Threshold merger), STRAT$\rightarrow$STRAX, MAID$\rightarrow$eMAID, HT$\rightarrow$HTX, REP$\rightarrow$REPv2, AGIX$\rightarrow$FET (ASI merger), XZC$\rightarrow$FIRO, TOMO$\rightarrow$VIC, NPXS$\rightarrow$PUNDIX, BNX redenomination, and DGD, whose dissolution refunded the ETH treasury to holders: a death with δ near 1, the mirror image of LUNA.

B. REGRESSIONS AND ROBUSTNESS

B.1 The Lindy test. Poisson regression of monthly death indicators on log age, offset $\log(1/12)$, standard errors clustered by asset. Lindy (hazard $= k/\text{age}$) requires a log-age coefficient of -1.

The log-survival–log-age slope over ages 1–10 runs from -0.21 (strict) to -0.49 (curated), i.e. a cumulative Pareto constant $\hat{k} \approx 0.2$–0.5 across definitions, an order of magnitude below the 5.6 the age rule would require. Two further robustness checks address the standard identification worry that an age effect might be a disguised cohort effect (age, cohort, and period are collinear, so only two can enter at once): replacing

Death definition	Deaths	γ_{age}, no FE	γ_{age}, year FE
Strict (feed termination)	234	+0.56** (0.07)	+0.44** (0.06)
Broad (feed + price)	263	+0.69** (0.07)	+0.56** (0.07)
Abandonment (all mechanical)	588	–	+0.51** (0.04)
Curated (final)	**520**	–	+0.42** **(0.04)**
Curated, 2025 feed-wall removed	–	–	+0.65** (0.08)
Curated, birth-cohort FE in place of year FE	520	–	+1.62** (0.10)
Elite (ever top-20) subsample	36	–	+0.68** (0.21)

Standard errors in parentheses, clustered by asset. ** significant at the 1% level; * at the 5% level. Every coefficient differs from the Lindy value of -1 at the 1% level.

calendar-period fixed effects with birth-cohort fixed effects *raises* the age elasticity to +1.62 (s.e. 0.10), so the rejection of Lindy is not an artifact of the period controls, and the period-FE +0.42 is the conservative figure. Parametrically, a Weibull fit to the curated durations gives shape $\rho = 1.57$ (95% CI 1.47–1.67), where $\rho > 1$ means a hazard rising with age and Lindy requires $\rho < 1$; a log-logistic fit (slightly preferred by AIC, 3,342 vs 3,371) implies a hump-shaped hazard that rises through the first decade of life. The implied median lifetime of a core-universe cryptoasset is 6.4 years under both the Weibull and the Kaplan–Meier estimates.

B.2 The concentration quantity is not a timing signal. Forward h-year log drift regressed on the real-time concentration quantity $x = \sigma^2 \cdot (\text{cap}/\text{participating wealth})$ and the age-rule regressor $z = \ln(1+5/A)/5$; heteroskedasticity- and autocorrelation-robust (HAC) standard errors with $h+6$ lags; overlapping monthly windows from 2012 (independent observations ≈ 2–4, so read t-statistics as descriptive).

Horizon	x alone	z alone	Horse race: x	Horse race: z	R^2 (joint)
3y	−45.1 (t −5.9)	+6.0 (t 5.0)	−39.1	+5.1	0.67
5y	−33.9 (t −4.1)	+7.0 (t 13.5)	−30.1	+6.5	0.81

The concentration quantity enters with the *wrong sign* as a predictor (market capitalization in its numerator makes it spike at mania tops) while the age regressor replicates the companion paper's coefficient (≈ 5–7, the power-law exponent). This is why the paper treats the marginal-holder condition as a date-by-date equilibrium restriction and never as a forecasting variable; conversely, it is also why the age rule's in-sample fit should not be read as evidence of a priced mechanism.

B.3 The companion paper's own metric, applied to the required-return path. A reader might ask the required-return path to pass the backtest that selected the age rule. It does not, and should not. Evaluated as a five-year forecast over the 2012–2021 windows, the survey-anchored lower edge of the band underpredicts realized returns by 65 points a year on average and is beaten by the age rule in 81% of windows; the band midpoint is nearly unbiased over the sample (realized exceeds it by 8 points a year on average) yet is still beaten in 83% of windows, because it does not track the timing of the surprises. That is the wedge, measured a third way: over the sample in which the age rule was fitted, realized returns exceeded any defensible contemporaneous required return by tens of points a year. A rate can be the correct required return and a poor forecast of a sample dominated by surprises; the backtest cannot distinguish an unbiased forecast from a reduced form that happens to track surprise decay.

B.4 Wealth-path robustness. The back-cast's wealth-per-participant ramp (a quarter of the 2021 level in 2012, 1.4 times by 2026) is a judgment about who held Bitcoin. Replacing it with the growth of US

household net worth per adult (Federal Reserve Z.1, about \$80T in 2013 to \$170T in 2026), so that early holders are assumed as wealthy as the average American rather than poorer, lowers the early back-cast weight and the band's lower edge: 2013 becomes 12–231% (baseline 17–231%), while 2017 (22–67%) and 2021 (21–58%, the anchor) are unchanged and 2026 becomes 9–25% (baseline 9–23%). All qualitative statements survive; the Section 5 envelope and its propagated interval use the wider w bounds.

B.5 Propagating input uncertainty. The 9–22% envelope of Section 5 sums the extremes of four input ranges: beta 0.33–1.05 on a 4% equity premium; risk aversion $\gamma \in [1.7, 2.8]$; holder weight $w \in [0.08, 0.19]$; mortality term 0.3–3 points; with the risk-free rate at 5.0% and $\sigma = 0.42$. To obtain a distribution rather than an envelope we draw each input independently and uniformly over its range, 400,000 draws, and compute the required return $r_f + \beta\,\mathrm{ERP} + \gamma\sigma^2 w + \lambda(1-\delta)$ for each. The median is 14.7%, the central 68% interval 12.9–16.7%, and the 90% interval 11.8–18.0%. The uniform assumption is deliberately uninformative; any distribution concentrated toward the interior of the ranges tightens the result. Narrowing the mortality term to 0.3–1.5 points, which is where the jump point estimate and a partial allowance for slow fades put it, moves the median to 13.9% and the 90% interval to 11.3–17.0%. Narrowing the holder weight to 0.12–0.19, the upper half supported by the market-clearing back-cast of Section 6, gives a median of 14.8% and a 90% interval of 12.4–17.4%. The same procedure gives the gold figures of Appendix C (beta uniform on $[-0.1, 0.1]$, σ on $[0.13, 0.20]$, w on $[0.02, 0.10]$, mortality on 0–0.3 points: median 5.5%, 90% interval 5.0–6.0%) and Bitcoin's digital-gold endpoint ($\sigma = 0.15$, w on $[0.02, 0.10]$, beta and mortality at Bitcoin's ranges: median 9.7%, 90% interval 7.8–11.6%). The script is `23_uncertainty.py` in the replication package.

B.6 Death-definition sensitivities. Volume-collapse deaths require an absolute floor (\$250k/day trailing volume) in addition to the 1%-of-peak rule; without the floor, 2021-mania base effects falsely kill operating mid-caps. The elite hazard is 3.54% with all curated deaths, 3.15% excluding the four contested zombies, and 2.36% (CI 1.51–3.51%) treating all twelve 2025 deaths as censoring; the Lindy rejection holds under every variant (B.1).

C. EXTENSIONS AND FORWARD-LOOKING HAZARDS

This appendix collects three applications of equation (1) that support the argument of the main text without being part of it: the condition applied to gold, the return meaning of "digital gold," and the treatment of hazards that have not yet fired.

C.1 Gold, as a check on the equation. Gold is the natural comparison, both because Bitcoin is so often called digital gold and because gold is the one other large asset that pays no income, so the whole of its return must come from price. Equation (1) prices it with the same four inputs, every one of which sits at the opposite end of the range from Bitcoin's. Market risk: gold's correlation with equities is close to zero over long samples (Erb and Harvey 2013), so the beta term is a few tenths of a point either way ($\beta \in [-0.1, 0.1]$). Concentration: gold is the most widely held asset on earth, about 224,000 tonnes above ground, of which 44% is jewelry, 18% central-bank reserves, and 23% private bars, coins, and ETFs (World Gold Council 2026), worth about \$29 trillion at today's price, some 6% of global household wealth; its marginal holder is a diversified investor with a few percent in it, not an enthusiast with a fifth. With volatility of 13–20% (13.1% over 1836–2011 in Barro and Misra 2016; higher in the last two years), a holder weight of 2–10%, and $\gamma \in [1.7, 2.8]$, the concentration term is 0.1–1.1 points, median 0.3. Mortality: gold has no failure mode that

destroys the asset; the closest analogues are sovereign restrictions, such as the United States' compulsory purchase of private gold in 1933 at the old official price before the dollar was devalued, which are jurisdiction risks to a holder rather than deaths of the asset, and we allow 0–0.3 points. The condition returns a required return for gold of about 5.5% a year (90% interval 5.0–6.0%; envelope 4.7–6.8%; Appendix B.5): the risk-free rate plus half a point, a Sharpe ratio near zero. This is a reproduced result, and that is its value here: Barro and Misra (2016) find that gold's real price appreciation averaged 1.1% a year over 175 years, with a covariance with consumption indistinguishable from zero, and conclude that its expected return should sit close to the risk-free rate; the equation reaches the same place from observable portfolios rather than from a consumption model. Gold's recent history then reads as Bitcoin's did in Section 6. From the end of 2023 to November 2026 gold compounded at about 27% a year (from about $2,060 to about $4,040 an ounce, with a gain of some 65% in 2025 alone and a flat twelve months since; LBMA 2026), roughly twenty points a year above anything its holders could have required. That excess is a wedge of the same kind as Bitcoin's: central-bank purchases, fiscal and geopolitical news, and ETF inflows the market had not priced, not a payment for risk. In equilibrium the equation puts gold's expected return from here back at 5–6% nominal; extrapolating 2024–25 would be the gold version of the age rule.

C.2 What "digital gold" would mean. The comparison also fixes the phrase in return terms. If Bitcoin's volatility fell to gold's 15% and its holder base diluted to gold's few percent per holder, the concentration term would all but vanish, as gold's has, and what remained would be the two terms gold does not carry: a beta of 0.33–1.05 and a mortality term of 0.3–3 points. The condition then gives about 10% a year (90% interval 8–12%; Appendix B.5). Only if those two also converged to gold's, with Bitcoin as uncorrelated with equities as gold and as immune to failure, would its required return reach gold's 5–6%. Digital gold, in other words, names a required return near the risk-free rate, the end state of the compression this paper documents, and the opposite of a 32% asset.

C.3 Forward-looking hazards the graveyard cannot see: the quantum threat. The hazard measured here is a backward-looking frequency, and some risks have never yet fired for anyone. The clearest is quantum computing: a cryptographically relevant quantum computer running Shor's algorithm would break the elliptic-curve signatures that secure Bitcoin, and the event would be exactly what the marginal-holder condition calls a jump, a λ-event with its own loss fraction. The model therefore *covers* the quantum threat by construction; what the graveyard cannot do is price it, because the realized death count contains no quantum events. Two observations discipline it. First, the loss given the event is scenario-dependent, not automatically total: roughly a quarter of Bitcoin's supply sits in addresses with exposed public keys, including the presumed-Satoshi coins (Deloitte's standing estimate is about 4 million BTC), and is the immediately vulnerable part, while a telegraphed advance in quantum hardware would allow a migration to post-quantum signatures that preserves most other value; so δ ranges from near zero (a surprise break) to substantial (an orderly transition). Second, if the market prices a material quantum hazard, it must appear where all jump risk appears, in deep out-of-the-money put prices. The option-implied crash intensity is therefore the running referee between the graveyard's 0.3% and any larger forward-looking hazard, quantum included; at this writing it shows no such wedge. An investor who privately assigns, say, a 1% annual probability to a disorderly quantum break with half of value lost should simply add 0.5 points to the mortality term; the equation accepts the view without modification.

D. CONSTRUCTING THE HOLDER WEIGHT

The holder weight w is the input the paper measures least directly, so this appendix shows how it is built (Figure 8). One number is measured: in 2021Q3 US households that owned any crypto held 19% of their financial assets in it on average (Candia, Coibion, Gorodnichenko, and Weber 2026), with the Austrian 2023 survey (OeNB 2025) as a cross-check. Everything else follows from market clearing, capitalization $= \bar{w}\times$ participating wealth, run backward and forward from that anchor.

D.1 Participation. The participant count is the share of US adults holding crypto in household surveys: 0.4% (2014) and 1.2% (2018) from the Federal Reserve's payment-choice surveys, 11% in 2021Q3 from Candia et al., 12–13% since from the Fed's SHED and Pew. The path is interpolated log-linearly between survey dates; the Coin Metrics count of addresses holding a balance supplies the shape only before the first survey point. Panel (a) also shows the alternative of scaling the address count to the 2021 point, which overstates participation five-fold in 2018 (6.4% against 1.2%) because addresses multiply faster than people; it is not used.

D.2 Wealth behind the holders. Participating wealth is anchored at capitalization divided by 0.19 in 2021Q3 (about $4.3 trillion) and moved with the participant count times wealth per participant, a ramp from a quarter of the 2021 level in 2012 to 1.4 times by 2026 (Appendix B.4 replaces the ramp with US net worth per adult). From 2024 a second class of holders enters through the spot ETFs and public-company treasuries, about 2.5 million BTC by mid-2026 (roughly 1.25 million in US spot ETFs and 1.26 million in corporate treasuries, Strategy alone about 0.85 million), worth about $190 billion at the May 2026 price. Their investors are not "crypto owners" in household surveys and hold small weights; at 1.5% of wealth they represent about $13 trillion of participating wealth, at 3% about $6 trillion, against $7 trillion for survey-type holders (panel b).

D.3 The weight. Dividing capitalization by participating wealth gives the implied average weight (panel c): about 3% in 2013, 17% in 2017, 18% in 2021, and in 2026 either 21% for survey-type holders alone or 8% once the institutional channel is counted at 1.5%. The path spikes at the 2013, 2017 and 2021 peaks because capitalization moves with the price while participation is smooth; that is the feature Appendix B.2 exploits as a negative predictor and the reason the paper treats the weight as an input to a date-by-date restriction rather than a signal. The band used in Section 6 runs from the diluted back-cast (lower edge) to the all-in holder of the option bound (0.9, upper edge); the 2026 range of Section 5, $w \in [0.08, 0.19]$, is the diluted value up to the surveyed value. Two omissions work in opposite directions and are left as they are: coins that are permanently lost (estimates of 3–4 million BTC) should be removed from capitalization, which would lower every weight by 15–20%, and the wealth of the roughly 15% of owners who hold more than half their portfolio in crypto is understated by an average, which would raise the weight of the marginal holder if she is drawn from that group.

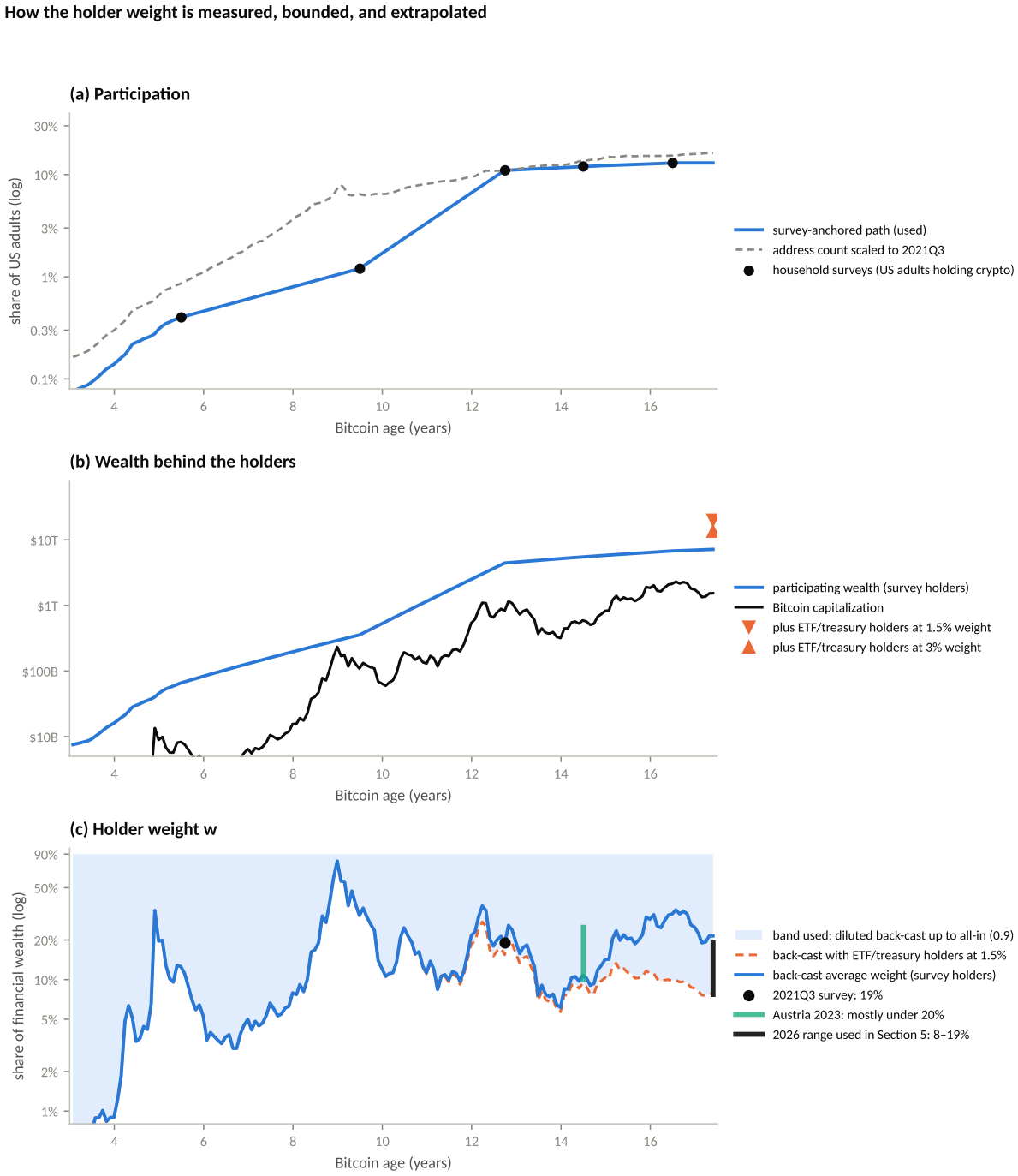


Figure 8: Constructing the holder weight. (a) Share of US adults holding crypto: survey points, the log-linear path used, and the address-count alternative not used. (b) Participating wealth of survey-type holders, Bitcoin's capitalization, and the 2026 additions for ETF and treasury investors at 1.5% and 3% weights. (c) The implied average weight, the 2021Q3 survey value, the Austrian cross-check, the diluted back-cast, and the band used in Section 6; all panels on logarithmic scales.